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Vortex-induced vibration mitigation in long-span bridges using a nonlinear energy sink inerter: Theoretical framework and application

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ABSTRACT

Long-span bridges, with their exceptionally low modal frequencies, are prone to wind-induced vibrations, notably vortex-induced vibrations (VIVs). Conventional linear dynamic vibration absorbers like tuned mass dampers (TMDs), and nonlinear dynamic absorbers such as nonlinear energy sinks (NESs), often struggle to mitigate VIV effectively at low frequencies due to excessive static displacements and limited achievable mass ratios. To overcome these challenges, this study proposed a novel vibration control device—a quasi-zero-stiffness nonlinear energy sink inerter (QZS-NESI). By introducing a positive linear stiffness component, the QZS-NESI achieves a quasi-zero stiffness configuration that compensates for static loads while maintaining low restoring forces. Meanwhile, the inclusion of the inerter provides a mass amplification effect, effectively reducing the static displacement requirement. The complexification-averaging (CX-A) technique was used to obtain the slow-flow dynamic model and steady-state dynamic model of the original system, which approximated by the CX-A method has been demonstrated to possess accuracy comparable to that of the original system. Moreover, this method enabled the direct determination of the steady-state amplitude and phase difference of the dynamic system, thereby elucidating the bifurcation structure of the system under varying parameters. The mechanical damping and VIV-related aerodynamic damping were identified as critical parameters significantly influencing the emergence of unstable branches. The Xihoumen Bridge, with a main span of 1650 m, was chosen as the reference bridge to evaluate the VIV control efficacy of the NESI system. Findings underscored the exceptional capacity of NESI for displacement reduction and broadband vibration mitigation.

1. Introduction

As wind-sensitive structures, bridge structures are prone to vibrations under wind loads, including flutter (Wang et al., 2023), buffeting (Cui et al., 2022), and vortex-induced vibrations (VIVs) (Ge et al., 2022). Various aerodynamic countermeasures such as wind fairings (Li et al., 2018), traffic barriers (Bai et al., 2020), guide plates (Zhang et al., 2020b), and porous grid plates (Wang et al., 2021a) have been widely employed to suppress VIVs in bridges. These aerodynamic measures achieve the reduction of wind speed passing through the girder and even the alteration of vortex shedding patterns between the air-fluid and the bridge through aerodynamic optimization to the girder configuration, thereby weakening the self-excited forces acting upon the bridge deck. The sensitivity to wind-induced vibrations, especially VIVs, is significantly strengthened as the modal frequency and damping ratio of long-span bridge structures decrease with the increasing span. In such cases, a single aerodynamic measure may not suffice, often necessitating a combination of multiple aerodynamic measures to eliminate wind-induced vibrations, thereby posing significant challenges to bridge maintenance and operation (Ge et al., 2022). Moreover, such a combination approach typically requires wind tunnel tests to be conducted before finalizing, thereby requiring a significant investment in costs during the wind-resistant design phase.

Mechanical control measures are considered as alternative effective approaches to address wind-induced vibrations, particularly VIVs. Such control strategies are capable of supplementing additional modal

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damping of the primary structure and facilitating vibration reduction, as exemplified by tuned mass dampers (TMDs), which was first proposed by Frahm Hermann (1911). Its outstanding vibration absorption performance and straightforward installation characteristics have led to its adoption in the VIV control of various bridge structures, such as the Rio–Niteroi Bridge (Battista Ronaldo and Pfeil Michèle, 2000), the Trans-Tokyo Bay Bridge (Fujino and Yoshida, 2002), the Chong Qi Bridge (Meinhardt et al., 2014), and the Humen Bridge. However, the effectiveness of TMDs is constrained by the necessity for their frequencies to be precisely matched with the modal frequencies of structural vibrations, thereby determining their vertical static displacements according to the well-known relationship $\Delta = g/\left(4\pi^2 f_s^2\right)$, where f_s is the frequency of bridge. This implies that for vertical VIVs with frequencies below 0.22Hz, the static displacement of the TMDs would exceed 5 m, often surpassing the height of the girder, making it challenging for the TMD to be utilized in suppressing VIVs. For long-span bridges with main spans exceeding 1000 m, the first vertical mode frequency typically remains below 0.2 Hz (Fujino and Dionysius, 2013). For instance, the Xihoumen Bridge, with a main span of 1650 m, has been identified to experience multiple instances of VIVs at a frequency of 0.1831 Hz (Li et al., 2011; Li et al., 2014). The low modal frequencies of VIVs pose a difficulty to the control of VIVs by TMDs in long-span bridge structures. To address this challenge, Xu et al., 2019 implemented a combination of tuned mass damper with inerter (TMDI) for mitigating wind-induced vibrations in long-span bridges. The significant reduction in static displacement was identified by adjusting the placement of the inerter within the TMD system, confirming the applicability of TMDI for wind-induced control in long-span bridge structures (Dai et al., 2021; Xu et al., 2020; Xu et al., 2022). Another critical factor constraining the utilization of TMD in long-span bridges is its exceptionally narrow frequency band (Aly, 2014), a feature that cannot be significantly enhanced by the inerter. Certainly, the frequency of the primary structure often fluctuates due to various uncertain factors such as aging, temperature variations, and extreme loading conditions. Observations from field monitoring of long-span bridges reveal fluctuations in modal frequencies and damping characteristics over different seasons (Tan et al., 2024). This variability suggests potential degradation in the performance of TMDS due to frequency discrepancies.

In 2001, Gendelman et al. (2001) and Vakakis and Gendelman (2001) discovered that nonlinear dynamic vibration absorbers with cubic stiffness could achieve transient resonance capture (TRC) with different frequency ratios to the primary structure. They achieved vibration suppression through irreversible one-way energy transfer, known as targeted energy transfer (TET). Hence, these absorbers are also referred to as nonlinear energy sink (NES). The presence of nonlinear stiffness in NES means that it does not have characteristic frequencies, which enables broadband energy absorption, thereby addressing the low robustness inherent in traditional linear absorbers (e.g., TMD). Both experimental and numerical studies have been carried out to demonstrate the robustness of NESs against changes or randomness in the natural frequencies of primary structures (McFarland et al., 2005; Saeed et al., 2022; Wang et al., 2021b). In recent years, cubic stiffness-type NES (Dai et al., 2017; Mehmood et al., 2014) and rotational-type NES (Chen et al., 2021; Franzini, 2021) have been utilized for mitigating VIVs in circular cylinders. Theoretical analyses and numerical simulation have both demonstrated their effectiveness in mitigating VIVs. In an extension of VIV mitigation, Yu et al., 2022 delved into the efficacy of VIV control in bridge structures employing nonlinear energy sink inerter (NESI) devices. This investigation underscored the inerter's installation position in influencing the performance of VIV control.

It is worth noting that for NES systems used to control vertical VIVs, the weight-induced nonlinear restoring force cannot be compensated for by static displacement (Chen et al., 2019). The weight leads to variations in stiffness at the equilibrium point of the NES system. NES systems designed without considering the weight effect exhibit performance degradation when employed for vertical vibration control (Li et al., 2022; Zhao et al., 2025). This is attributed to the alteration in the vibration frequency components of the NES system under the influence of weight, leading to a decreased efficiency in resonance capture between the NES and the primary structure. In this study, the authors propose a quasi-zero-stiffness nonlinear energy sink inerter (QZS-NESI), in which the inclusion of positive linear stiffness components offsets the gravitational weight of the mass block, thereby achieving a quasi-zero stiffness configuration. This design is expected to reduce static displacement and eliminate the influence of weight on the nonlinear stiffness characteristics of the NESI system. Given that the mechanisms governing the VIV mitigation capabilities of NESI devices featuring self-balancing weight functionalities in the context of bridge applications remain obscure, and a comprehensive theoretical framework outlining these principles remains conspicuously absent. Hence, this study endeavors to formulate a theoretical framework pertaining to the utilization of this enhanced NESI in mitigating VIV in long-span bridges. This endeavor seeks to elucidate the bifurcation characteristics inherent in the VIV control capabilities of the enhanced NESI, thereby furnishing a robust theoretical underpinning for its implementation. The objective is to achieve static displacement reduction and wide frequency band control, thereby realizing stronger robustness of the mechanical mitigation system.

2. Modelling of bridge coupled QZS-NESI system

2.1. Definition of problem

Fig. 1(a) depicts a widely recognized undamped NES system, which is composed of a set of symmetric linear springs (McFarland et al., 2005). Due to geometric nonlinearity, when the oscillator moves vertically, the restoring force is nearly cubic with respect to displacement. Consequently, the static equilibrium position under the gravity meets the following relationship under Taylor expansion:

$$k_N \Delta^3 \cong mg \tag{1}$$

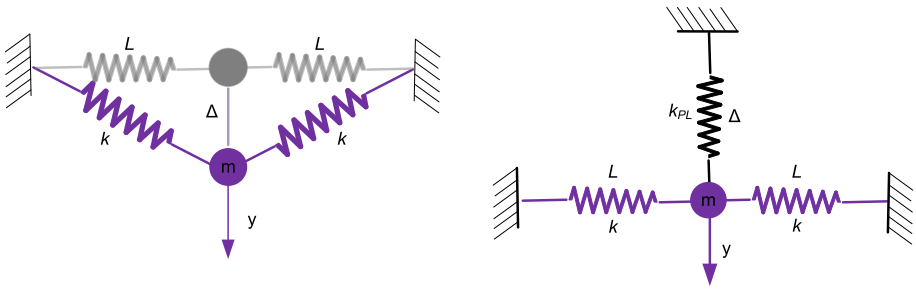
where $k_N = \frac{k}{2l^2}$ is cubic stiffness coefficient; Δ is static displacement induced by weight; m is the mass of oscillator; g is gravitational acceleration; At time t, the dynamic equilibrium condition of the system is expressed as follows:

$$m\ddot{y} + k_N\left(y^3 + 3y^2\Delta + 3y\Delta^2\right) = 0 \tag{2}$$

where y is dynamic displacement; The static displacement induced by the weight results in two additional nonlinear restoring forces ($k_N(3y^2\Delta + 3y\Delta^2)$) in the system, which cannot be compensated. This indicates that any variation in the mass will have a significant impact on the dynamic displacement, thereby introducing uncertainty into vibration mitigation. To prevent the static displacement caused by the weight from influencing the nonlinear restoring forces of the NES system, a quasi-zero-stiffness mechanism is introduced by incorporating a positive linear stiffness element, as illustrated in Fig. 1(b). Once the static displacement induced by the weight is balanced by the positive linear stiffness element, the laterally symmetric springs are incorporated into the system, leading to the following dynamic equilibrium relationship:

$$m\ddot{y} + k_{PL}y + k_N y^3 = 0 \tag{3}$$

where k_{PL} is the positive linear stiffness, which is used to compensate weight of oscillator, i.e., $k_{PL}\Delta = mg$. In this modified QZS-NESI system, the effect of weight on the nonlinear restoring forces is eliminated. It is important to note that the linear stiffness must be carefully designed to preserve the pronounced nonlinear characteristics.



(a) NES system (b) QZS-NES system

Fig. 1. Vertical static equilibrium state under weight.

2.2. Generalized physical model

The proposed NESI incorporates a positive linear stiffness (k_{PL}) element to balance the weight of the mass blocks, forming the QZS-NESI system. This system effectively eliminates the impact of weight effects on VIV mitigation. VIV mitigation is achieved through frequency modulation facilitated by a set of transverse springs forming the cubic nonlinear stiffness (k_N). An inerter device with mass amplification effect (apparent mass b) is attached at one end to the mass block and at the other end to the bridge deck. The configuration of the QZS-NESI system in a large-span bridge is illustrated in Fig. 2, and it comprises stiffness elements (k_{PL} and k_N), damping elements (c_N), mass blocks (m_N), and an inerter (b).

Consistent with Chen et al., 2022, the cable and girder of the suspension bridge are represented as a tensioned beam. Here, the cable supplies the tension force T, while the girder contributes the bending stiffness EI. The per unit mass of the tensioned beam is denoted as m_s, with L representing the overall length of the suspension bridge. Hence, the original system depicted in Fig. 2 can be simplified as shown in Fig. 3. The vertical displacement at the location where the inerter is connected to the bridge deck is represented by $y(e,t)$, while the vertical displacement at the location where the mass block is connected to the bridge deck is represented by $y(d,t)$. The vertical displacement of the mass block is denoted by v.

2.3. Theoretical model of coupled dynamic system

Given the complete balancing of the weight by the linear stiffness element, the dynamic equations of the coupled system with external load $F(x,t)$ remain consistent before and after static equilibrium. They can be formally expressed as

$$\begin{aligned} &EIy_{xxxx}(x,t)+C_s y_t(x,t)+m_s y_{tt}(x,t)-Ty_{xx}(x,t)+\{k_{PL}[y(d,t)-v(t)]\\ &\quad+k_N[y(d,t)-v(t)]^3+C_N[y_t(d,t)-\dot{v}(t)]\}\delta(x-d)\\ &\quad+b[y_{tt}(e,t)-\ddot{v}(t)]\delta(x-e)=F(x,t) \end{aligned} \tag{4}$$

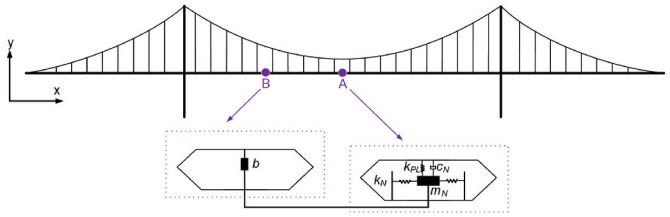


Fig. 2. Long-span bridge with QZS-NESI system.

$$\begin{aligned} &m_N\ddot{v}(t)-k_{PL}[y(d,t)-v(t)]-k_N[y(d,t)-v(t)]^3-C_N[y_t(d,t)-\dot{v}(t)]\\ &\quad-b[y_{tt}(e,t)-\ddot{v}(t)]=0 \end{aligned} \tag{5}$$

where the short-hand notation for partial differentiation is enforced e.g., $(\cdot)_{xx}\equiv\partial^2(\cdot)/\partial x^2$, $(\cdot)_t\equiv\partial(\cdot)/\partial t$, ….; C_s is the damping coefficient of the primary structure; $v(t)$ is the dynamic displacement of the mass block; δ is Dirac function.

The transverse displacement $y(x,t)$ can be decomposed into the summation of modal displacements as

$$y=\sum_{r=1}^{\infty}\phi_r(x)a_r(t) \tag{6}$$

$$\phi_r(x)=\sin(r\pi x/L) \tag{7}$$

where $\phi_r(x)$ is the rth-order mode shape of the beam; $a_r(t)$ is the rth-order modal coordinate of the beam. Eq. (6) is substituted into Eqs. (4) and (5), resulting in

$$\begin{aligned} EI\ &\sum_{r=1}^{\infty}\frac{d^4\phi_r(x)}{dx^4}a_r(t)+C_s\sum_{r=1}^{\infty}\phi_r(x)\dot{a}_r(t)+m_s\sum_{r=1}^{\infty}\phi_r(x)\ddot{a}_r(t)\\ &\quad-T\sum_{r=1}^{\infty}\frac{d^2\phi_r(x)}{dx^2}a_r(t)+\left\{k_{PL}\left[\sum_{r=1}^{\infty}\phi_r(d)a_r(t)-v(t)\right]\right.\\ &\quad+k_N\left[\sum_{r=1}^{\infty}\phi_r(d)a_r(t)-v(t)\right]^3\\ &\quad\left.+C_N\left[\sum_{r=1}^{\infty}\phi_r(d)\dot{a}_r(t)-\dot{v}(t)\right]\right\}\delta(x-d)\\ &\quad+b\left[\sum_{r=1}^{\infty}\phi_r(e)\ddot{a}_r(t)-\ddot{v}(t)\right]\delta(x-e)=F(x,t) \end{aligned} \tag{8}$$

$$\begin{aligned} &m_N\ddot{v}(t)-k_{PL}\left[\sum_{r=1}^{\infty}\phi_r(d)a_r(t)-v(t)\right]-k_N\left[\sum_{r=1}^{\infty}\phi_r(d)a_r(t)-v(t)\right]^3\\ &\quad-C_N\left[\sum_{r=1}^{\infty}\phi_r(d)\dot{a}_r(t)-\dot{v}(t)\right]-b\left[\sum_{r=1}^{\infty}\phi_r(e)\ddot{a}_r(t)-\ddot{v}(t)\right]=0 \end{aligned} \tag{9}$$

where $\phi_r(d)$ presents the value of mode shape at QZS-NESI position; $\phi_r(e)$ is the value of modal shape at inerter coupling beam position. Considering the orthogonality of modes, they meet the following relationships

$$\int_0^L\phi_r(x)\phi_p(x)dx=0, r\neq p \tag{10}$$

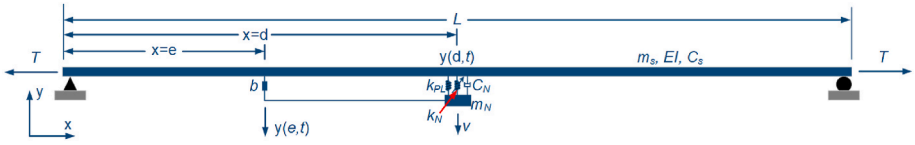


Fig. 3. Tensioned beam with QZS-NESI system.

$$\int_0^L m_s\phi_r(x)\phi_p(x)dx = m_p\delta_{rp}, r, p = 1, 2, \ldots \tag{11}$$

$$\int_0^L EI\frac{\partial^2\phi_r}{\partial x^2}\frac{\partial^2\phi_p}{\partial x^2}dx - \int_0^L T\frac{d\phi_r}{dx}\frac{d\phi_p}{dx}dx = m_p\omega_p^2\delta_{rp}, r, p = 1, 2, \ldots \tag{12}$$

Multiplying any modal shapes $\phi_p(x)$ by Eqs (8) and (9), and subsequently integrating along the length (L) of the beam, results in

$$\begin{aligned} m_p\left[\omega_p^2 a_p(t) + 2\omega_p\xi_p\dot{a}_p(t) + \ddot{a}_p(t)\right] + \left\{k_{PL}\left[\sum_{r=1}^{\infty}\phi_r(d)a_r(t) - v(t)\right]\right. \\ + k_N\left[\sum_{r=1}^{\infty}\phi_r(d)a_r(t) - v(t)\right]^3 + C_N\left[\sum_{r=1}^{\infty}\phi_r(d)\dot{a}_r(t) - \dot{v}(t)\right]\Bigg\}\phi_p(d) \\ + b\left[\sum_{r=1}^{\infty}\phi_r(e)\ddot{a}_r(t) - \ddot{v}(t)\right]\phi_p(e) = \int_0^L F(x,t)\phi_p(x) \end{aligned} \tag{13}$$

$$\begin{aligned} m_N\ddot{v}(t) - k_{PL}\left[\sum_{r=1}^{\infty}\phi_r(d)a_r(t) - v(t)\right] - k_N\left[\sum_{r=1}^{\infty}\phi_r(d)a_r(t) - v(t)\right]^3 \\ - C_N\left[\sum_{r=1}^{\infty}\phi_r(d)\dot{a}_r(t) - \dot{v}(t)\right] - b\left[\sum_{r=1}^{\infty}\phi_r(e)\ddot{a}_r(t) - \ddot{v}(t)\right] = 0 \end{aligned} \tag{14}$$

This system can be non-dimensionalized by introducing the following normalized parameters and variables

$$\begin{aligned} \tau = \omega_p t, V = \frac{v}{D}, A = \frac{a}{D}, \xi_p = \frac{C_s}{2m_p\omega_p}, \lambda = \frac{C_N}{2m_p\omega_p}, S_l = \frac{k_{PL}}{m_p\omega_p^2}, S_n = \frac{k_N D^2}{m_p\omega_p^2}, \mu \\ = \frac{m_N}{m_p}, \beta = \frac{b}{m_p}, l = \frac{x}{L}, Q(\tau) = \frac{L\int_0^1 F(l,\tau)\phi_p(lL)dl}{m_p\omega_p^2 D} \end{aligned} \tag{15}$$

where τ is the dimensionless time; V is the displacement of nonlinear oscillator normalized by a characteristic length D, which is the height of beam structure in this study; A is the pth-order modal coordinate normalized by a characteristic length D; ξ_p and λ are the dimensionless damping coefficients of beam and QZS-NESI, respectively. S_l and S_n are normalized stiffness coefficients of the QZS-NESI; μ is the mass ratio of the QZS-NESI; β is the inertance ratio of the QZS-NESI; l is spanwise coordinate normalized by length of beam $l = \frac{x}{L}$; $Q(\tau)$ is the normalized external load. The dimensionless expression for vortex-excited force of the simplified Scanlan model is as follows:

$$Q(\tau) = \frac{U_r}{2\pi m_r}\int_0^1\left[Y_1\left(1 - \varepsilon\phi_p^2(lL)A_p^2(\tau)\right)\right]\phi_p^2(lL)\dot{A}_p(\tau)dl \tag{16}$$

where Y_1 and ε are aerodynamic parameters; U_r is the reduced wind speed and m_r is mass ratio characterizing the ratio of modal mass of the primary structure to the mass of the air fluid. These two parameters are defined as

$$U_r = \frac{2\pi U}{\omega_p D}, m_r = \frac{m_p}{\rho D^2 L} \tag{17}$$

Hence, The dimensionless forms of Eqs. (13) and (14) under single-mode VIV are

$$\begin{aligned} A_p(\tau) - \left[\eta_1 - \eta_2 A_p^2(\tau)\right]\dot{A}_p(\tau) + \ddot{A}_p(\tau) + \left\{S_l\left[\phi_p(d)A_p(\tau) - V(\tau)\right]\right. \\ + S_n\left[\phi_p(d)A_p(\tau) - V(\tau)\right]^3 + 2\lambda\left[\phi_p(d)\dot{A}_p(\tau) - \dot{V}(\tau)\right]\Big\}\phi_p(d) \\ + \beta\left[\phi_p(e)\ddot{A}_p(\tau) - \ddot{V}(\tau)\right]\phi_p(e) = 0 \end{aligned} \tag{18}$$

$$\begin{aligned} \mu\ddot{V}(\tau) - S_l\left[\phi_p(d)A_p(\tau) - V(\tau)\right] - S_n\left[\phi_p(d)A_p(\tau) - V(\tau)\right]^3 - 2\lambda\left[\phi_p(d)\dot{A}_p(\tau)\right. \\ \left. - \dot{V}(\tau)\right] - \beta\left[\phi_p(e)\ddot{A}_p(\tau) - \ddot{V}(\tau)\right] = 0 \end{aligned} \tag{19}$$

where η_1 and η_2 are Van der Pol parameters, which are defined as

$$\eta_1 = \frac{U_r}{4\pi m_r}\int_0^1 Y_1\phi_p^2(lL)dl - 2\xi_p, \eta_2 = \frac{U_r}{4\pi m_r}\int_0^1 Y_1\varepsilon\phi_p^4(lL)dl \tag{20}$$

From Eqs. (18) and (19), it can be observed that the vibration response of the system is influenced by parameters η_1 and η_2 related to vortex-excited forces, the stiffness coefficients S_l and S_n of the QZS-NESI system, damping coefficient λ, mass ratio u, inertance ratio β, as well as the modal values $\phi_p(d)$ and $\phi_p(e)$.

3. Steady-state response model

3.1. Slow flow dynamic model

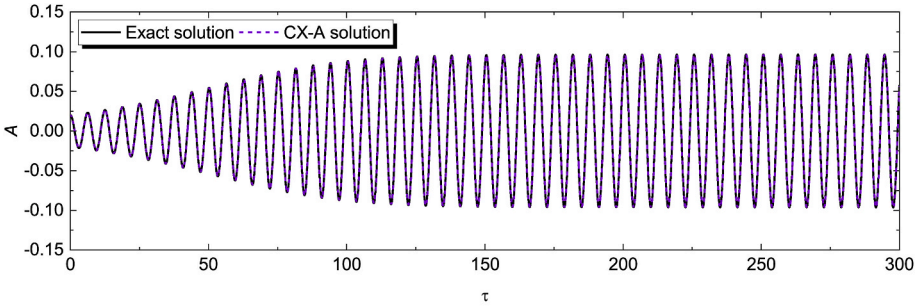
To analytically study the transient and steady-state motions of the beam coupled QZS-NESI system under VIVs, the complexification-averaging (CX-A) technique proposed by Manevitch Leonid (1999) is used. According to the CX-A technique, complex variables representing the motion of the primary structure and QZS-NESI are introduced:

$$\psi_1 = \dot{A} + jA = \varphi_1 e^{j\tau}, \psi_2 = \dot{V} + jV = \varphi_2 e^{j\tau} \tag{21}$$

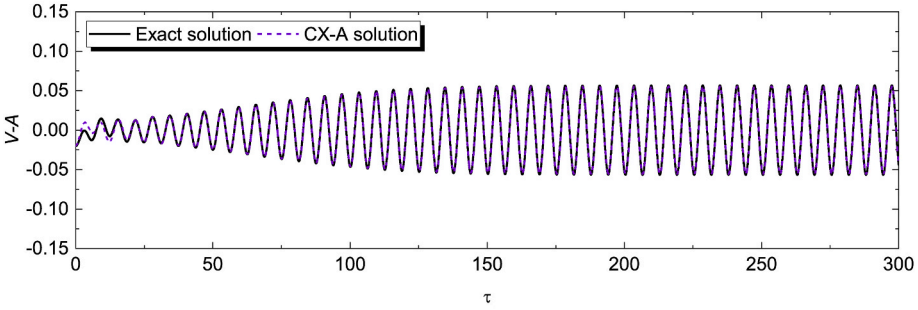
where $j^2 = -1$; φ_1 and φ_2 denote the slowly-varying complex amplitudes of the primary structure and QZS-NESI, respectively; $e^{j\tau}$ is the fast-varying components at the (linearized) natural frequency of the system. The motion of the primary structure and QZS-NESI can be formally represented using the above complex variables as follows:

$$\begin{aligned} A = \frac{\varphi_1 e^{j\tau} - \varphi_1^* e^{-j\tau}}{2j}, \dot{A} = \frac{\varphi_1 e^{j\tau} + \varphi_1^* e^{-j\tau}}{2}, \ddot{A} = \dot{\varphi}_1 e^{j\tau} + \frac{j}{2}\left(\varphi_1 e^{j\tau} - \varphi_1^* e^{-j\tau}\right) V \\ = \frac{\varphi_2 e^{j\tau} - \varphi_2^* e^{-j\tau}}{2j}, \dot{V} = \frac{\varphi_2 e^{j\tau} + \varphi_2^* e^{-j\tau}}{2}, \ddot{V} = \dot{\varphi}_2 e^{j\tau} + \frac{j}{2}\left(\varphi_2 e^{j\tau} - \varphi_2^* e^{-j\tau}\right) \end{aligned} \tag{22}$$

By expressions Eq. (22), transient dynamics are divided into slow and fast components. That is, the transient responses of the system described by Eq. (18) and Eq. (19) consist of rapid oscillations modulated by slowly varying envelopes. The averaging process is applied to all high-frequency components higher than unity, yielding a set of two complex-valued modulation equations used to govern the slow-flow

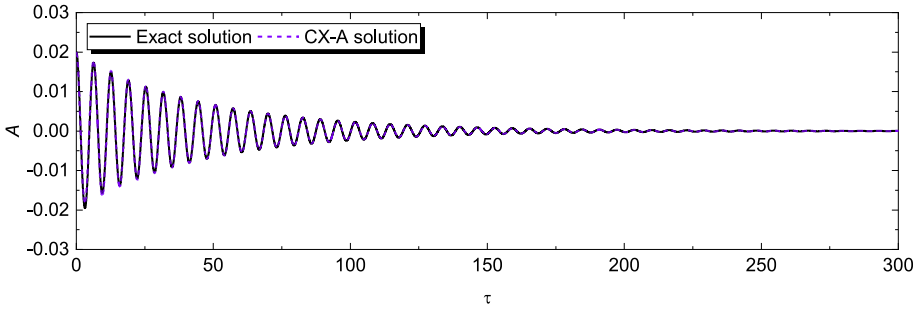


(a) Displacement of primary structure

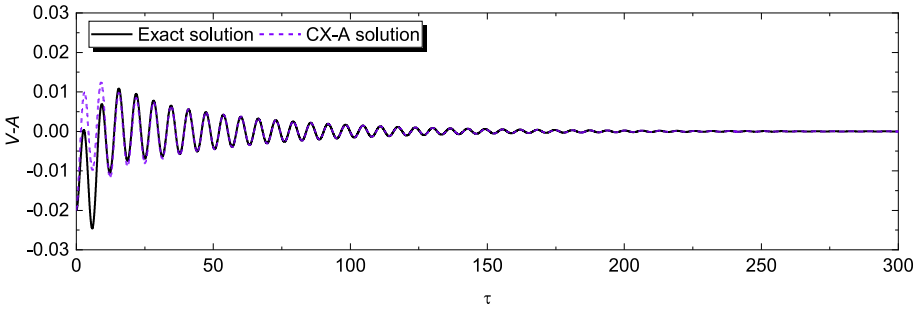


(b) Displacement of QZS-NESI

Fig. 4. Survival VIV ($\mu = 0.001$, $\beta = 0.1$, $S_l = 0.01$, $S_n = 10$, $\lambda = 0.01$, $\eta_1 = 0.05$, $\eta_2 = 18.75$, and $\phi_e = 0.6$); initial conditions are given by $(A(0), \dot{A}(0), V(0), \dot{V}(0)) = (0.02, 0, 0, 0)$.



(a) Displacement of primary structure



(b) Displacement of QZS-NESI

Fig. 5. Eliminated VIV ($\mu = 0.001$, $\beta = 0.4$, $S_l = 0.01$, $S_n = 100$, $\lambda = 0.1$, $\eta_1 = 0.05$, $\eta_2 = 18.75$, and $\phi_e = 0.3$); initial conditions are given by $(A(0), \dot{A}(0), V(0), \dot{V}(0)) = (0.02, 0, 0, 0)$.

dynamics. Considering that the vibration absorber is typically arranged at the location of maximum modal value ($\phi_d = 1$), the slow flow dynamical model is expressed as

$$\left(1+\beta\phi_e^2-\frac{\beta^2\phi_e^2}{\mu+\beta}\right)\dot{\varphi}_1+\left(\frac{j}{2}\beta\phi_e^2-\frac{\eta_1}{2}\right)\varphi_1-\frac{j}{2}\beta\phi_e\varphi_2+\frac{\eta_2}{8}\varphi_1|\varphi_1|^2 +\frac{\beta\phi_e}{\mu+\beta}\left\{\frac{j}{2}(\mu+\beta)\varphi_2-\frac{j}{2}\beta\phi_e\varphi_1\right\}+\left(1-\frac{\beta\phi_e}{\mu+\beta}\right)\left\{\left(\lambda-j\frac{S_l}{2}\right)(\varphi_1-\varphi_2) -j\frac{3S_n}{8}(\varphi_1-\varphi_2)|\varphi_1-\varphi_2|^2\right\}=0 \tag{23}$$

$$\left(\mu+\beta-\frac{\beta^2\phi_e^2}{1+\beta\phi_e^2}\right)\dot{\varphi}_2+\frac{j}{2}(\mu+\beta)\varphi_2-\frac{j}{2}\beta\phi_e\varphi_1+\frac{\beta\phi_e}{1+\beta\phi_e^2}\left\{\left(\frac{j}{2}\beta\phi_e^2-\frac{\eta_1}{2}\right)\varphi_1 -\frac{j}{2}\beta\phi_e\varphi_2+\frac{\eta_2}{8}\varphi_1|\varphi_1|^2\right\}-\left(1-\frac{\beta\phi_e}{1+\beta\phi_e^2}\right)\left\{\left(\lambda-j\frac{S_l}{2}\right)(\varphi_1-\varphi_2) -j\frac{3S_n}{8}(\varphi_1-\varphi_2)|\varphi_1-\varphi_2|^2\right\}=0 \tag{24}$$

In Figs. 4 and 5, the validation of the averaged systems by comparing the averaged responses to the (numerically) exact solutions is examined. The comparison is demonstrated in scenarios where the VIV persists following the integration of the QZS-NESI as well as in instances where it is effectively suppressed. The adequacy of the approximations to the original dynamics provided by the averaged systems Eqs. (23) and (24) are confirmed through the comparisons depicted in Figs. 4 and 5. A minor deviation in the initial stage is observed solely in the scenario where the VIV is eradicated within the QZS-NESI system (Fig. 5(b)). In this instance, the precise QZS-NESI response incorporates an extra rapid, high-frequency component, which is filtered out during the averaging process. Nevertheless, the displacement evolution of the primary structure in the averaged system closely matches that of the original system, indicating that the TRC frequency ratio for suppressing the VIV should be 1:1.

3.2. Steady-state dynamic evolution

The steady-state response is generally characterized by both amplitude and phase. Thus, in this subsection, the steady-state dynamic models are developed based on the established slow-flow dynamic models. For simplification of the derivation process, the slow flow dynamic models Eq. (23) and Eq. (24) are equivalently expressed as:

$$\dot{\varphi}_1=\left(-ja_1+a_2-a_3|\varphi_1|^2\right)\varphi_1-\left(a_4-ja_5-ja_6|\varphi_1-\varphi_2|^2\right)(\varphi_1-\varphi_2) \tag{25}$$

$$\dot{\varphi}_2=-jb_1\varphi_2+\left(-jb_2+b_3-b_4|\varphi_1|^2\right)\varphi_1 +\left(b_5-jb_6-jb_7|\varphi_1-\varphi_2|^2\right)(\varphi_1-\varphi_2) \tag{26}$$

where coefficients a_i $(i = 1, 2\ldots6)$ and b_i $(i = 1, 2\ldots7)$ in the equation represent system parameters, with their detailed expressions provided in Appendix A.

The slowly-varying complex amplitudes are expressed in polar form

$$\varphi_i=\alpha_i e^{j\theta_i}, \alpha_i, \theta_i \epsilon R, i=1,2 \tag{27}$$

$$\Phi\equiv\theta_1-\theta_2 \tag{28}$$

where α_i and θ_i represent the slowly-varying amplitude and phase of each degree of freedom, respectively; Φ is the phase difference between the oscillations of the primary structure and the QZS-NESI. Substituting them into Eqs. (25) and (26), and performing Euler transformation, yields the following forms:

$$\begin{aligned}\dot{\alpha}_1\cos\theta_1-\alpha_1\dot{\theta}_1\sin\theta_1+j\dot{\alpha}_1\sin\theta_1+j\alpha_1\dot{\theta}_1\cos\theta_1&=-ja_1\alpha_1\cos\theta_1\\&+a_1\alpha_1\sin\theta_1+a_2\alpha_1\cos\theta_1+ja_2\alpha_1\sin\theta_1-a_3\alpha_1^3\cos\theta_1\\&-ja_3\alpha_1^3\sin\theta_1-a_4\alpha_1\cos\theta_1+a_4\alpha_2\cos\theta_2-ja_4\alpha_1\sin\theta_1\\&+ja_4\alpha_2\sin\theta_2+ja_5\alpha_1\cos\theta_1-ja_5\alpha_2\cos\theta_2-a_5\alpha_1\sin\theta_1\\&+a_5\alpha_2\sin\theta_2+ja_6\left(\alpha_1^2+\alpha_2^2-2\alpha_1\alpha_2\cos\Phi\right)(\alpha_1\cos\theta_1-\alpha_2\cos\theta_2)\\&-a_6\left(\alpha_1^2+\alpha_2^2-2\alpha_1\alpha_2\cos\Phi\right)(\alpha_1\sin\theta_1-\alpha_2\sin\theta_2)\end{aligned} \tag{29}$$

$$\begin{aligned}\dot{\alpha}_2\cos\theta_2-\alpha_2\dot{\theta}_2\sin\theta_2+j\dot{\alpha}_2\sin\theta_2+j\alpha_2\dot{\theta}_2\cos\theta_2&=-jb_1\alpha_2\cos\theta_2\\&+b_1\alpha_2\sin\theta_2-jb_2\alpha_1\cos\theta_1+b_2\alpha_1\sin\theta_1+b_3\alpha_1\cos\theta_1\\&+jb_3\alpha_1\sin\theta_1-b_4\alpha_1^3\cos\theta_1-jb_4\alpha_1^3\sin\theta_1+b_5\alpha_1\cos\theta_1\\&-b_5\alpha_2\cos\theta_2+jb_5\alpha_1\sin\theta_1-jb_5\alpha_2\sin\theta_2-jb_6\alpha_1\cos\theta_1\\&+jb_6\alpha_2\cos\theta_2+b_6\alpha_1\sin\theta_1-b_6\alpha_2\sin\theta_2-jb_7\left(\alpha_1^2\right.\\&\left.+\alpha_2^2-2\alpha_1\alpha_2\cos\Phi\right)(\alpha_1\cos\theta_1-\alpha_2\cos\theta_2)+b_7\left(\alpha_1^2+\alpha_2^2\right.\\&\left.-2\alpha_1\alpha_2\cos\Phi\right)(\alpha_1\sin\theta_1-\alpha_2\sin\theta_2)\end{aligned} \tag{30}$$

By setting the real and imaginary parts separately equal to zero, two sets of real modulation equations are obtained.

$$\begin{aligned}\dot{\alpha}_1\cos\theta_1-\alpha_1\dot{\theta}_1\sin\theta_1&=a_1\alpha_1\sin\theta_1+a_2\alpha_1\cos\theta_1-a_3\alpha_1^3\cos\theta_1\\&-a_4\alpha_1\cos\theta_1+a_4\alpha_2\cos\theta_2-a_5\alpha_1\sin\theta_1+a_5\alpha_2\sin\theta_2\\&-a_6\left(\alpha_1^2+\alpha_2^2-2\alpha_1\alpha_2\cos\Phi\right)(\alpha_1\sin\theta_1-\alpha_2\sin\theta_2)\end{aligned} \tag{31}$$

$$\begin{aligned}\dot{\alpha}_1\sin\theta_1+\alpha_1\dot{\theta}_1\cos\theta_1&=-a_1\alpha_1\cos\theta_1+a_2\alpha_1\sin\theta_1-a_3\alpha_1^3\sin\theta_1\\&-a_4\alpha_1\sin\theta_1+a_4\alpha_2\sin\theta_2+a_5\alpha_1\cos\theta_1-a_5\alpha_2\cos\theta_2\\&+a_6\left(\alpha_1^2+\alpha_2^2-2\alpha_1\alpha_2\cos\Phi\right)(\alpha_1\cos\theta_1-\alpha_2\cos\theta_2)\end{aligned}$$

$$\begin{aligned}\dot{\alpha}_2\cos\theta_2-\alpha_2\dot{\theta}_2\sin\theta_2&=b_1\alpha_2\sin\theta_2+b_2\alpha_1\sin\theta_1+b_3\alpha_1\cos\theta_1\\&-b_4\alpha_1^3\cos\theta_1+b_5\alpha_1\cos\theta_1-b_5\alpha_2\cos\theta_2+b_6\alpha_1\sin\theta_1\\&-b_6\alpha_2\sin\theta_2+b_7\left(\alpha_1^2+\alpha_2^2-2\alpha_1\alpha_2\cos\Phi\right)(\alpha_1\sin\theta_1-\alpha_2\sin\theta_2)\end{aligned} \tag{32}$$

$$\begin{aligned}\dot{\alpha}_2\sin\theta_2+\alpha_2\dot{\theta}_2\cos\theta_2&=-b_1\alpha_2\cos\theta_2-b_2\alpha_1\cos\theta_1+b_3\alpha_1\sin\theta_1\\&-b_4\alpha_1^3\sin\theta_1+b_5\alpha_1\sin\theta_1-b_5\alpha_2\sin\theta_2-b_6\alpha_1\cos\theta_1\\&+b_6\alpha_2\cos\theta_2-b_7\left(\alpha_1^2+\alpha_2^2-2\alpha_1\alpha_2\cos\Phi\right)(\alpha_1\cos\theta_1-\alpha_2\cos\theta_2)\end{aligned}$$

The steady-state dynamical model can be obtained by algebraically manipulating these two sets of equations as

$$\dot{\alpha}_1=\left(a_2-a_4-a_3\alpha_1^2\right)\alpha_1+a_4\alpha_2\cos\Phi-\left[a_5\alpha_2+a_6\alpha_2\left(\alpha_1^2+\alpha_2^2\right)\right]\sin\Phi +a_6\alpha_1\alpha_2^2\sin 2\Phi \tag{33}$$

$$\dot{\alpha}_2=\left[b_3\alpha_1-b_4\alpha_1^3+b_5\alpha_1\right]\cos\Phi-b_5\alpha_2+\left[b_2\alpha_1+b_6\alpha_1+b_7\alpha_1\left(\alpha_1^2+\alpha_2^2\right)\right]\sin\Phi -b_7\alpha_1^2\alpha_2\sin 2\Phi \tag{34}$$

$$\begin{aligned}\alpha_1\alpha_2\dot{\Phi}=(a_5-a_1-b_6+b_1)\alpha_1\alpha_2+\left(b_2\alpha_1^2+b_6\alpha_1^2-a_5\alpha_2^2\right)\cos\Phi\\-\left(a_4\alpha_2^2+b_3\alpha_1^2-b_4\alpha_1^4+b_5\alpha_1^2\right)\sin\Phi+\left(\alpha_1^2+\alpha_2^2\right.\\\left.-2\alpha_1\alpha_2\cos\Phi\right)\left[a_6\left(\alpha_1\alpha_2-\alpha_2^2\cos\Phi\right)+b_7\left(\alpha_1^2\cos\Phi-\alpha_1\alpha_2\right)\right]\end{aligned} \tag{35}$$

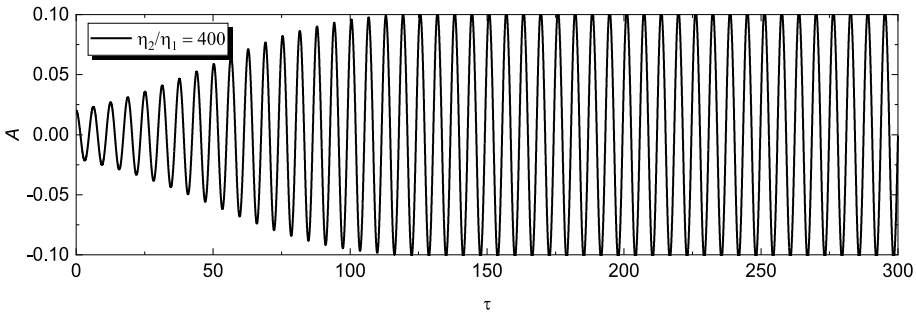
The steady-state dynamical models Eqs. (33)–(35) can be rewritten in the following compact form:

$$\dot{\alpha}_1=f_1(\alpha_1,\alpha_2,\Phi) \tag{36}$$

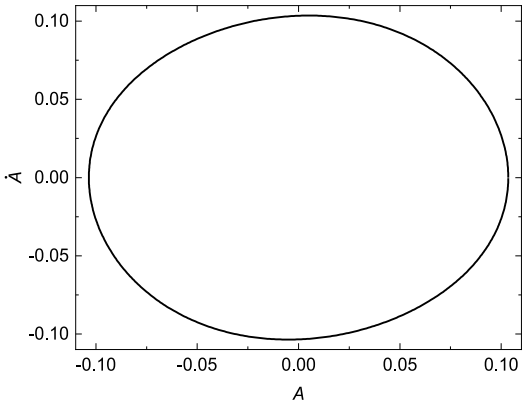
$$\dot{\alpha}_2=f_2(\alpha_1,\alpha_2,\Phi)$$

$$\alpha_1\alpha_2\dot{\Phi}=f_3(\alpha_1,\alpha_2,\Phi)$$

The steady-state amplitudes α_1, α_2, and the phase difference Φ are

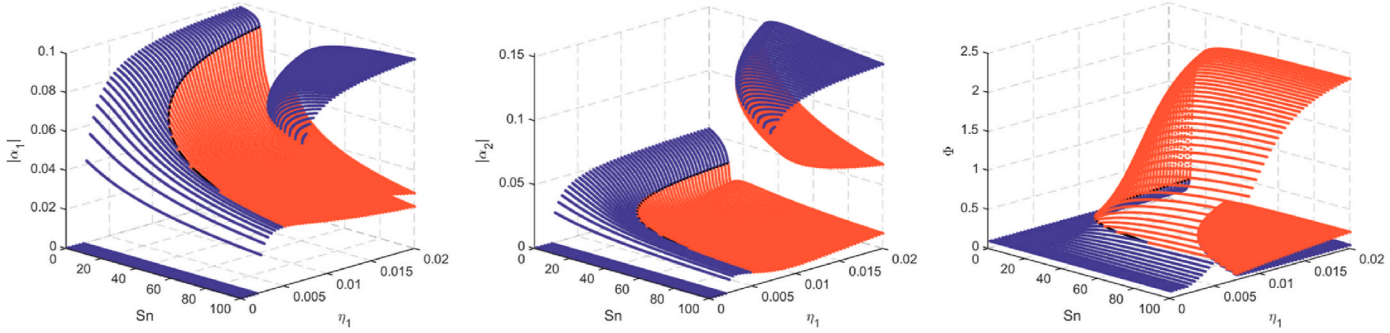


(a) VIV evolution



(b) Phase plane of VIV

Fig. 6. VIV of beam with aerodynamic parameters of $\eta_2/\eta_1 = 400$.



(a) LCO of primary structure (b) LCO of QZS-NESI (c) Phase difference

Fig. 7. Bifurcation diagram of system with respect to cubic stiffness S_n and aerodynamic parameter η_1

determined by solving the set of homogeneous nonlinear algebraic equations.

$$f_1(\alpha_1^*, \alpha_2^*, \Phi^*) = f_2(\alpha_1^*, \alpha_2^*, \Phi^*) = f_3(\alpha_1^*, \alpha_2^*, \Phi^*) = 0 \tag{37}$$

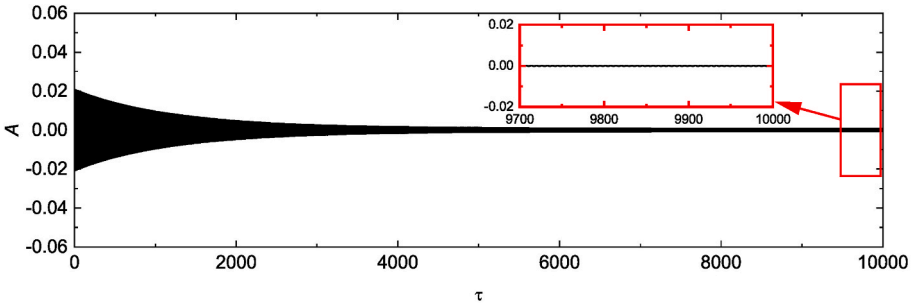
where α_1^*, α_2^* and, Φ^* are equilibrium points. The stability of each obtained equilibrium can be ascertained through the eigenvalues of the Jacobian matrix of Eqs. (33)–(35), which is provided in Appendix B. The nontrivial solutions of these nonlinear algebraic equations are difficult to obtain explicitly. Thus, the solution of steady-state equilibrium points and stability analysis is conducted by using the Matcont package proposed by Dhooge, Annick et al., 2003.

4. Bifurcation characteristics of beam coupled QZS-NESI system

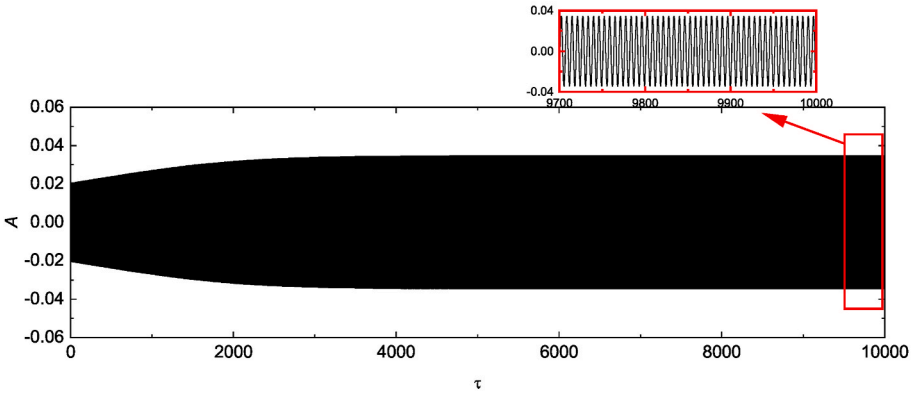
Given that the aerodynamic coefficients (η_1 and η_2), stiffness coefficients (S_l and S_n), damping coefficient (λ), mass parameters (u and β), and inerter position (ϕ_e) all influence the steady-state response of the system, the VIV bifurcation characteristics under variations in these parameters are observed in this section.

4.1. Effect of aerodynamic damping

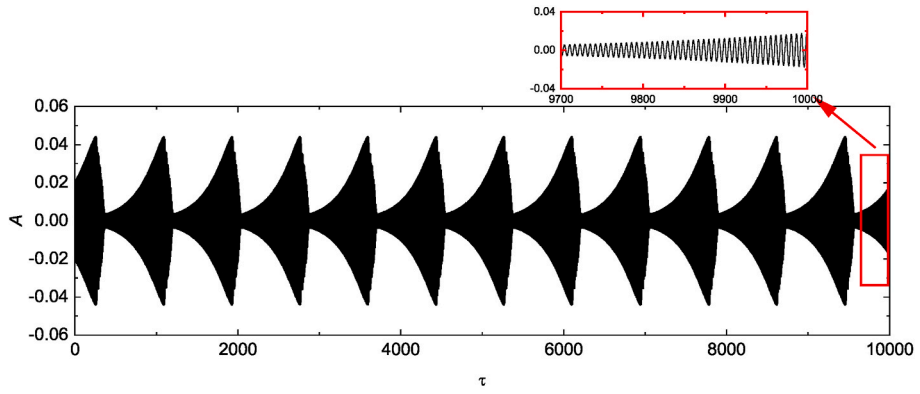
In the absence of damping in the primary structure, the relationship of VIV aerodynamic parameters is as follows:



(a) Eliminated LCO with $S_n = 50$ and $\eta_1 = 0.0005$

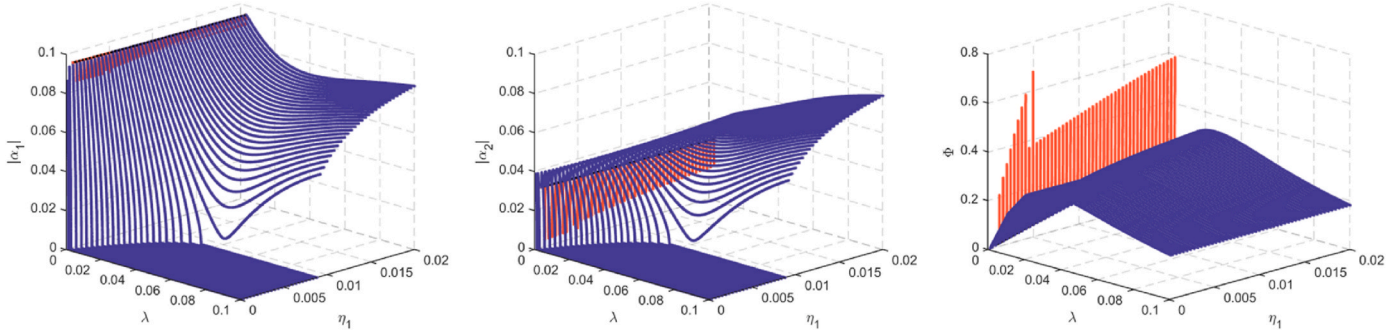


(b) Suppressed LCO with $S_n = 50$ and $\eta_1 = 0.003$



(c) SMR with $S_n = 50$ and $\eta_1 = 0.01$

Fig. 8. Evolution of VIV of primary structure corresponding to bifurcation structure.



(a) LCO of primary structure (b) LCO of QZS-NESI (c) Phase difference

Fig. 9. Bifurcation diagram of system with respect to mechanical damping λ and aerodynamic parameter η_1

$$\frac{\eta_2}{\eta_1} = \varepsilon\chi \tag{38}$$

where χ is modal factor, which is defined as $\chi = \frac{\int_0^1 \phi_p^4(lL)dl}{\int_0^1 \phi_p^2(lL)dl} = \frac{3}{4}$ in beam structure. The aerodynamic parameter ε significantly governs the amplitude of VIV limit cycle oscillation (LCO), with typically smaller ε corresponding to larger LCO amplitudes (Zhang, 2023). For a single-degree-of-freedom LCO oscillator in the beam structure, when $\varepsilon \approx 533$ ($\eta_2/\eta_1 = 400$), its VIV amplitude ratio is approximately 0.1, as shown in Fig. 6. The reported maximum VIV amplitude ratio of long-span bridge structures does not exceed 0.1 (Ge et al., 2022; Li et al., 2014). Hence, in this study, the VIV aerodynamic parameters for long-span bridges are determined to be $\eta_2/\eta_1 = 400$.

According to Eq. (20), without consideration of structural damping, the aerodynamic parameter η_1 is controlled by the reduced wind speed U_r, the mass ratio m_r, and the parameter Y_1. Table 1 presents the range of values for these relevant parameters during VIV of long-span bridges, from which the range of U_r, Y_1, and m_r can be determined as [6, 14], [6, 35], and [1000, 6500], respectively. Based on the parameter ranges mentioned above, the widest possible range for η_1 could be [2E-4, 0.02].

Fig. 7 depicts the bifurcation structure of the beam-coupled QZS-NESI system under fixed parameters $\mu = 0.001, \beta = 0.1, S_l = 0.01, \lambda = 0.005$, and $\phi_e = 0.6$, as the aerodynamic parameter η_1 and cubic stiffness S_n vary. All non-trivial curves in the bifurcation diagram represent periodic oscillations of the system, that is, LCOs. The blue lines represent stable branches, while the red lines denote unstable branches, and the black lines indicate Hopf bifurcation points. According to the bifurcation evolution of the primary structure (Fig. 7(a)), it is noted that when the aerodynamic damping parameter η_1 is less than 0.002, LCO is eliminated due to the presence of system damping. As η_1 increases, LCO becomes difficult to eliminate and is instead suppressed. The suppression effect of LCO increases with the increase in the stiffness coefficient S_n. When η_1 exceeds 0.005, a Hopf bifurcation occurs at higher stiffness coefficients, leading to an unstable branch. More importantly, As η_1 increases, the stiffness coefficient S_n corresponding to the appearance of the Hopf bifurcation point decreases, and the LCO amplitude increases, indicating a significant difficulty in suppressing LCO. An important observation is that once η_1 surpasses 0.008, the system's nonlinearity is intensified. With the increase in S_n, the system transits through the subcritical Hopf bifurcation zone, leading to the occurrence of abrupt changes in the LCO as the stiffness coefficient varies. In the QZS-NESI system, this amplitude jump phenomenon becomes more prominent. As shown in Fig. 7(b), with increasing stiffness, the stable amplitude of the oscillator jumps from 0.05 to 0.15, indicating insufficient robustness of LCO suppression. The stable phase difference amplitude is minimally affected by S_n and η_1, despite an increase in its unstable phase difference with rising η_1, as shown in Fig. 7(c).

The accuracy of the bifurcation structure illustrated in Fig. 7, obtained from the steady-state dynamical models Eqs. (33)–(35), can be confirmed by examining the displacement evolution of the original system Eq. (18) and Eq. (19) under initial conditions of $(A(0), \dot{A}(0), V(0), \dot{V}(0)) = (0.02, 0, 0, 0)$, as depicted in Fig. 8. To ensure obtaining the steady-state displacement of the primary structure, the dimensionless integration time (τ) is set to 10000. Fig. 8(a) illustrates the VIV evolution of the primary structure under a small value of η_1, where the LCO of the primary structure is eliminated, consistent with the results provided by the bifurcation diagram. At η_1 value of 0.003, consistent with the bifurcation structure, the LCO cannot be eliminated anymore, but is effectively suppressed, as shown in Fig. 8(b). When η_1 enters the unstable branches, although LCO is still suppressed, its amplitude fluctuates cyclically, and the system keeps a strongly modulated response (SMR) state (Vakakis et al., 2009, 2022), which is a continuous sequence of repetitive TRCs, involving resonance capture, escape, and recapture within the resonant manifold of the dynamics (Fig. 8(c)).

According to Fig. 7, it can be observed that irrespective of the variations in η_1, the system does not enter the subcritical Hopf bifurcation region when the stiffness coefficient S_n is less than 10. Fig. 9 depicts the bifurcation structure of the system as η_1 and λ change, with fixed parameters $S_l = 0.01$, $S_n = 10$, $\mu = 0.001$, $\beta = 0.1$, and $\phi_e = 0.6$. The occurrence of the Hopf bifurcation and the subsequent unstable branches in the system are only observed when the damping coefficient λ is extremely small (nearly zero). The LCO can be eliminated only when η_1 is less than 0.009; conversely, when η_1 exceeds 0.009, the LCO can only be suppressed rather than eliminated, as shown in Fig. 9(a) and (b). It is worth noting that as η_1 increases, the optimal damping coefficient (λ_{opt}) to eliminate the LCO increases, and conversely, it decreases when the LCO is suppressed. Specifically, with the increase of η_1, the optimal damping coefficient λ_{opt} for suppressing LCO decreases from 0.045 to 0.04. They differ slightly, which facilitates subsequent design applications. The phase difference of the system remains largely unaffected by η_1, while it increases as λ reaches its maximum value before declining (Fig. 9(c)).

4.2. Effect of linear stiffness

The linear stiffness elements in the QZS-NESI system play a crucial role in balancing the weight of system and thus constraining static displacement. The equilibrium relationship is

$$k_{PL}\Delta = (m_N + \alpha b)g \tag{39}$$

where Δ is static stroke induced by QZS-NESI system, and α is inertance coefficient indicating the ratio between the physical mass and apparent

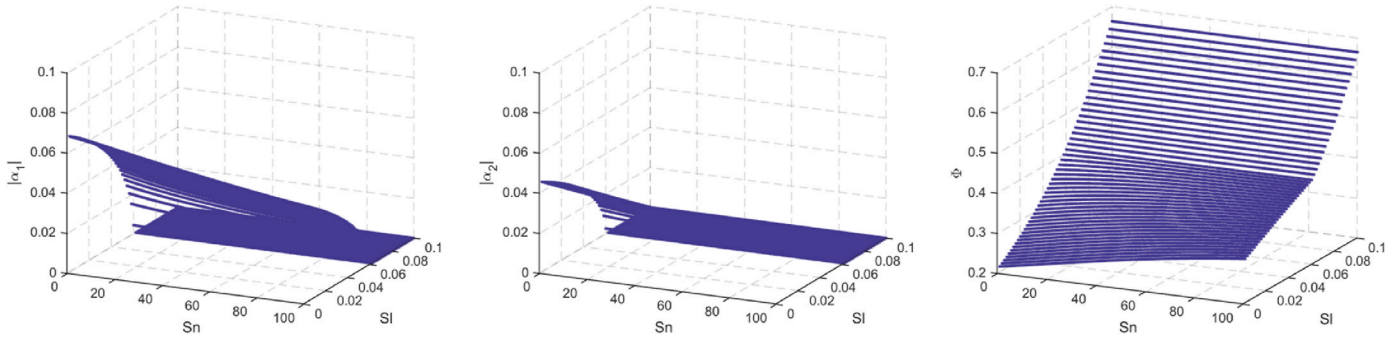


Fig. 10. Bifurcation diagram of system with respect to cubic stiffness S_n and linear stiffness S_l

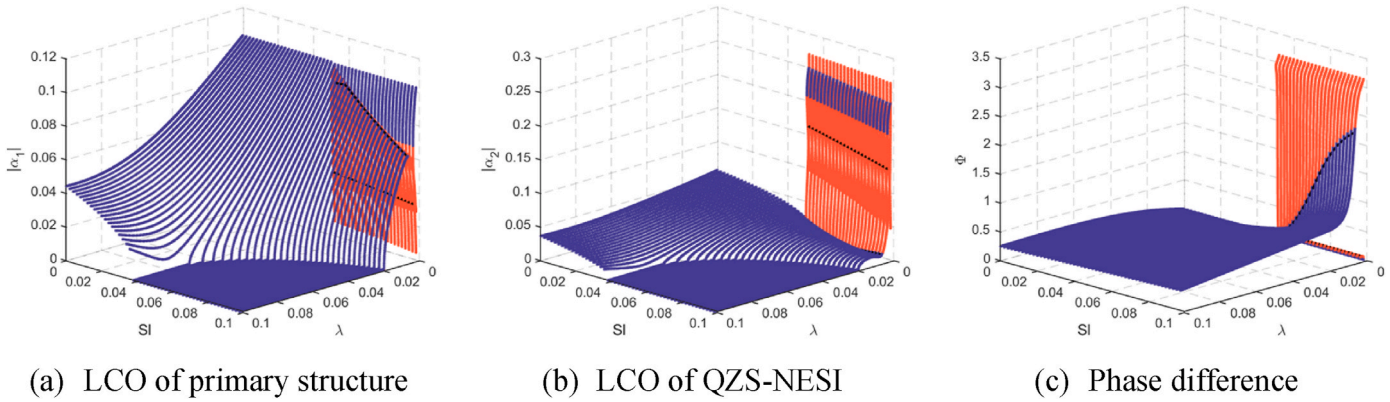


Fig. 11. Bifurcation diagram of system with respect to mechanical damping λ and linear stiffness S_l

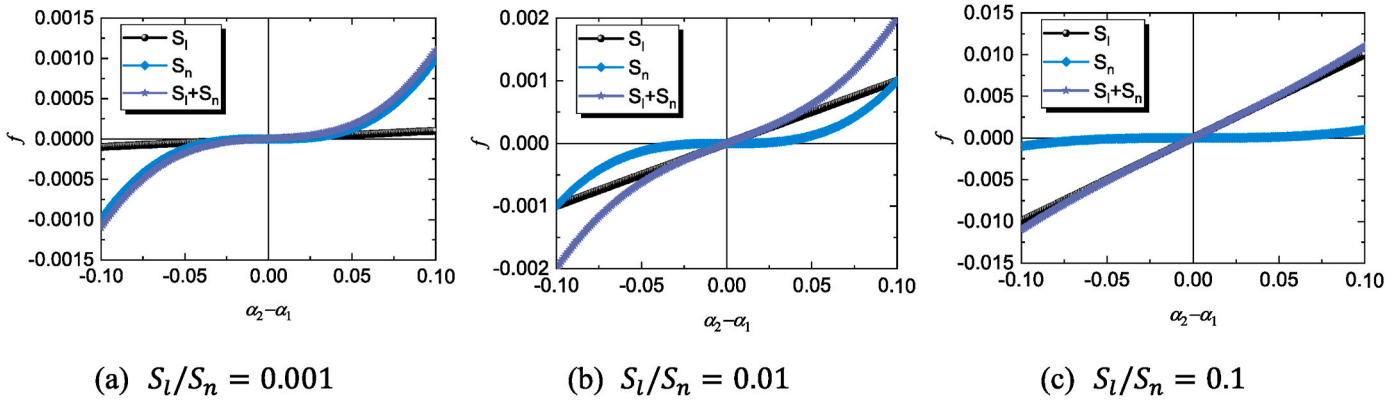


Fig. 12. Dimensionless restoring force versus relative displacement with different stiffness ratios.

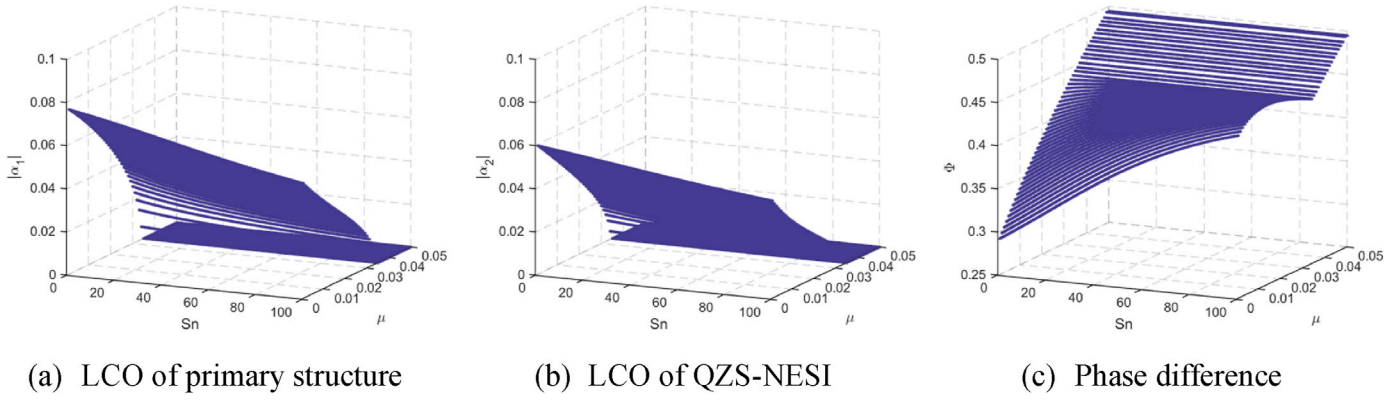


Fig. 13. Bifurcation diagram of system with respect to cubic stiffness S_n and mass ratio μ

mass of inerter. Substituting the dimensionless variables in Eq. (15) into Eq. (39) to yield

$$S_l\delta = (\mu + \alpha\beta)\gamma \tag{40}$$

where $\delta = \Delta/D$ is dimensionless static displacement, and $\gamma = g/\left(\omega_p^2 D\right)$ is dimensionless gravitational acceleration. According to (Lu et al., 2017), the potential range of α is [1/500 1/5000]. For long-span bridge, the range of γ is [0.2 8]. The ideal range for the static stroke δ should be [0.1 0.5], to allow space for mass block vibration. Hence, the range of linear stiffness coefficients S_l suitable for VIV mitigation in long-span bridges can be determined as [4E-4 0.1], with parameters of $\mu = 0.001$, and $\beta = 0.2$.

The bifurcation structure of the system varies with S_n and S_l under fixed parameters of $\eta_1 = 0.02$, $\lambda = 0.04$, $\mu = 0.001$, $\beta = 0.2$, and $\phi_e = 0.6$, as illustrated in Fig. 10. Within the potential range of linear stiffness S_l, increasing its value proves beneficial in mitigating LCO amplitude, with LCO being eliminated when S_l exceeds 0.06, as depicted in Fig. 10 (a) and (b). In this scenario, even when S_n is 0, LCO can still be eliminated due to the presence of S_l, which serves to capture vibrations. Additionally, the robustness is enhanced by the inerter, explaining why increasing S_l also results in the elimination of LCO. It is noteworthy that enhancing S_l is more efficient than augmenting S_n in the function of LCO elimination. In line with previous observations, the more pronounced the elimination of LCO, the greater the phase difference, corresponding to larger S_n and S_l (Fig. 10(c)). With the cubic stiffness S_n fixed at 10, an

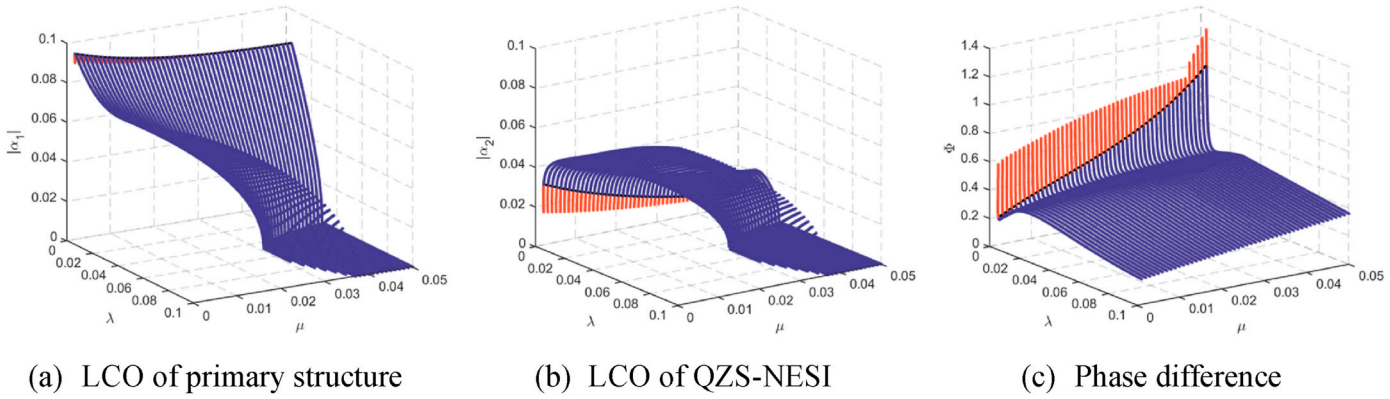


Fig. 14. Bifurcation diagram of system with respect to mechanical damping λ and mass ratio μ

increase in the linear stiffness S_l ($S_l > 0.05$) results in the appearance of unstable branches in the low damping region ($\lambda < 0.002$), as shown in Fig. 11. It can be observed that once λ reaches 0.07, LCO can be eliminated with S_l at 0.038. Furthermore, with larger S_l, the required λ to eliminate LCO can be smaller, as shown in Fig. 11(a) and (b). Under larger S_l values, the phase difference increases as λ decreases, and it significantly increases when λ is extremely small, eventually jumping to an unstable branch, as depicted in Fig. 11(c).

It is important to note that although the introduced linear stiffness S_l can eliminate weight effects and limit static displacement, it also affects the nonlinear restoring force, as shown in Fig. 12. Specifically, when the ratio of the linear stiffness coefficient to the nonlinear stiffness coefficient (S_l/S_n) is too small, the coupled nonlinear restoring force resembles that of a system without the linear stiffness element (Fig. 12(a)). However, this may lead to excessively large static displacement due to the weight. As the stiffness ratio increases further, the restoring force of the coupled system behaves linearly within a small deformation range and transitions to nonlinear as deformation increases, as illustrated in Fig. 12(b). This configuration maintains the broad-frequency vibration mitigation capability due to its intrinsic nonlinear stiffness characteristics. However, when the stiffness ratio becomes excessively large, the restoring force of the coupled system is primarily linear, lacking significant nonlinear characteristics (Fig. 12(c)). As a result, the system degenerates into a linear dynamic vibration absorber, leading to the loss of its broad-frequency vibration control capability. Hence, to ensure that the QZS-NESI system retains its broad-frequency vibration mitigation capability, the stiffness ratio should not exceed 0.01. In other words, the linear stiffness coefficient S_l should be at least 100 times smaller than the nonlinear stiffness coefficient S_n.

4.3. Effect of mass ratio

Fig. 13 depicts the bifurcation structure of the system as the mass ratio (μ) and stiffness coefficient (S_n) vary under fixed parameters of $\lambda = 0.04$, $\eta_1 = 0.02$, $S_l = 0.01$, $\beta = 0.1$, and $\phi_e = 0.6$. It can be observed that with the damping coefficient λ set to the optimal value as depicted in Fig. 9, the system no longer experiences Hopf bifurcations, rendering all stable branches across the entire parameter domain. The LCO is categorized into two states by the mass ratio μ: suppression and elimination, as shown in Fig. 13(a) and (b). Specifically, when $\mu < 0.035$, the LCO is suppressed; otherwise, it is eliminated. In the region where LCO is suppressed, increasing stiffness S_n aids in diminishing LCO amplitude, whereas in the region where LCO is eliminated, it can be achieved solely through linear stiffness S_l. Given that the selected aerodynamic damping parameter η_1 is already a potentially large value, in practical applications, the required μ for eliminating LCO can be smaller than 0.035. In contrast to LCO, the phase difference increases as μ and S_n increase (Fig. 13(c)). However, in the LCO elimination region, the phase difference is unaffected by changes in S_n.

When the aerodynamic damping coefficient η_1 is 0.02, with fixed parameters of $S_l = 0.01$, $S_n = 10$, $\beta = 0.1$, and $\phi_e = 0.6$, the mechanical damping coefficient λ needs to exceed 0.004 for the system to avoid entering the Hopf bifurcation region, as depicted in Fig. 14. As the mass ratio increases, the LCO undergoes a transition from suppression to elimination, with a critical mass ratio μ of 0.03, as shown in Fig. 14(a) and (b). In the region where LCO is suppressed, the optimal λ increases with μ, while when LCO can be eliminated, the minimum λ required for LCO elimination decreases with μ. The phase difference is hardly influenced by changes in μ, while it undergoes an evolution of initially increasing and then decreasing with λ (Fig. 14(c)), consistent with the observations in Fig. 9(c).

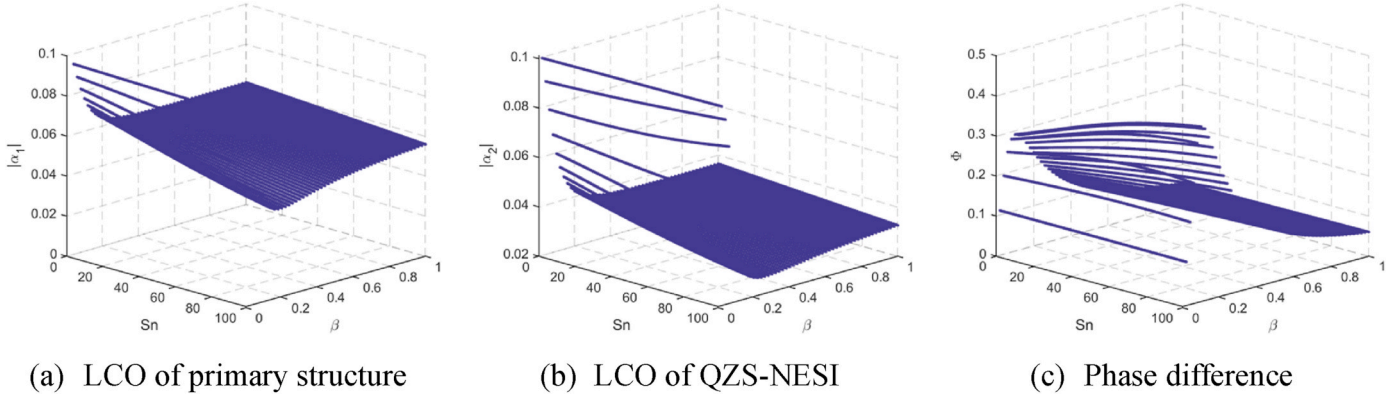


Fig. 15. Bifurcation diagram of system with respect to cubic stiffness S_n and inertance ratio β

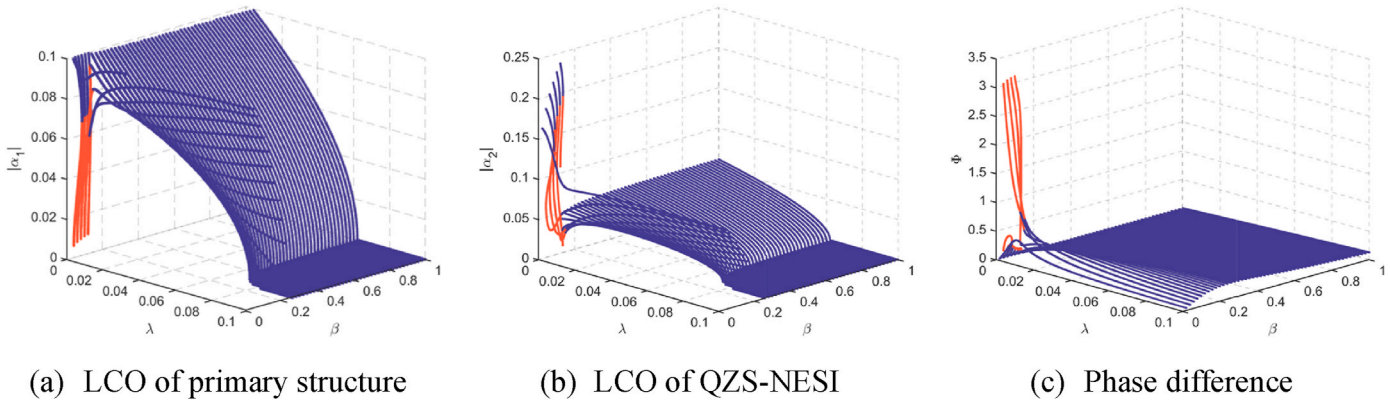


Fig. 16. Bifurcation diagram of system with respect to mechanical damping λ and inertance ratio β

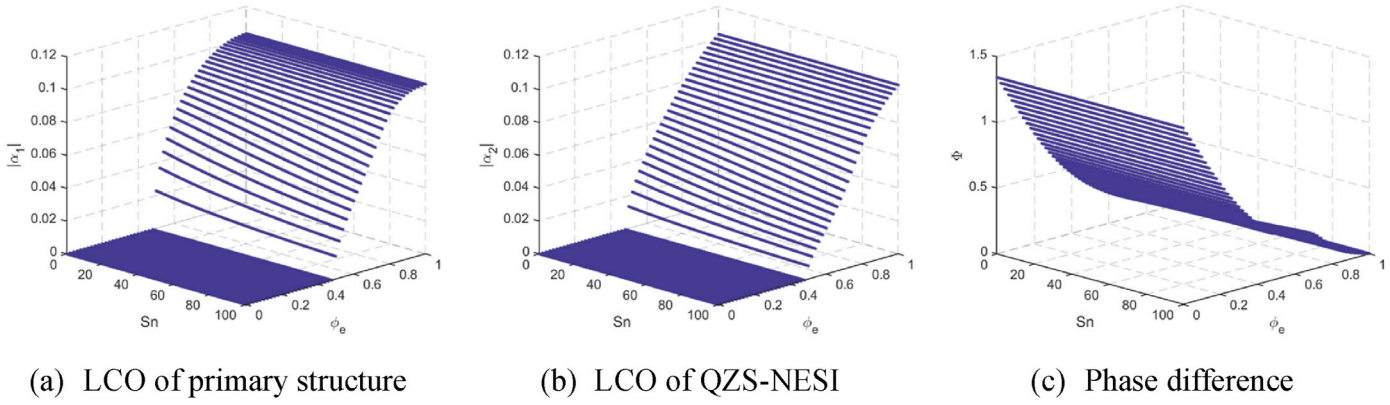


Fig. 17. Bifurcation diagram of system with respect to cubic stiffness S_n and modal value ϕ_e

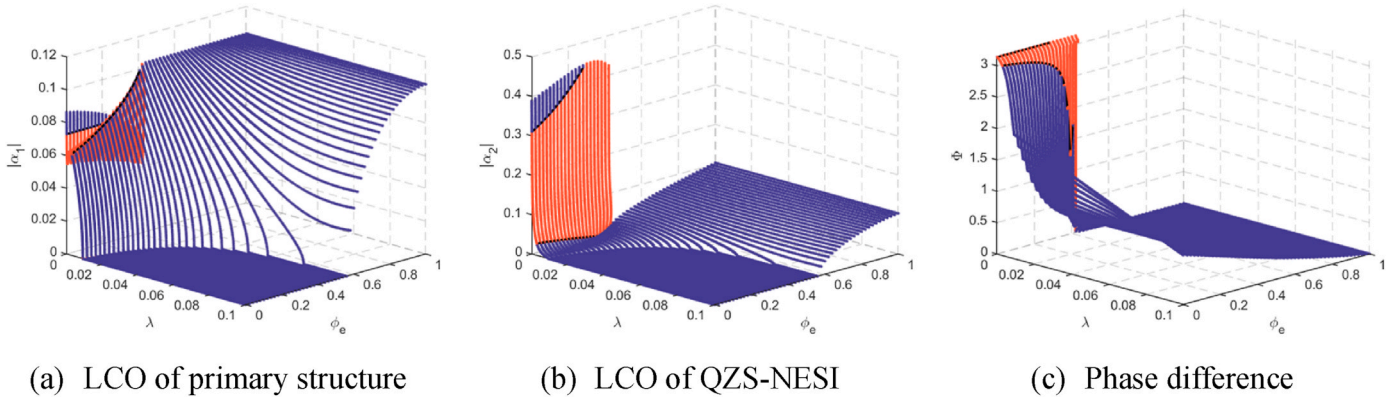


Fig. 18. Bifurcation diagram of system with respect to mechanical damping λ and modal value ϕ_e

4.4. Effect of inerter

In this subsection, the evolution of bifurcation characteristics in the system due to changes in the inertance ratio (β) and location (ϕ_e) of the inerter is identified.

4.4.1. Inertance ratio

The bifurcation evolution of the system under fixed parameters of $\eta_1 = 0.02$, $\lambda = 0.04$, $S_l = 0.01$, $\mu = 0.001$, and $\phi_e = 0.6$, with changes in S_n and β is depicted in Fig. 15. The reduction in LCO amplitude is facilitated by an increase in β, but once β exceeds 0.2, this reduction effect becomes ineffective, as illustrated in Fig. 15(a) and (b). Beyond this critical threshold, the reduction in LCO amplitude by increasing β is no longer observed. An important observation is that the phase difference reaches its maximum when the LCO amplitude is minimized, as depicted in Fig. 15(c). When the inertance ratio is small ($\beta < 0.12$), the system experiences unstable LCO with low mechanical damping λ ($\lambda < 0.01$) with fixed parameters of $\eta_1 = 0.02$, $S_l = 0.01$, $S_n = 10$, $\mu = 0.001$, and $\phi_e = 0.6$, as depicted in Fig. 16. When β is below 0.2, the optimal λ increases with β, and when β exceeds 0.2, LCO elimination can be achieved by increasing λ ($\lambda > 0.06$), Fig. 16(a) and (b). The phase difference remains nearly constant when β exceeds 0.2 (Fig. 16(c)).

4.4.2. Installation location

The installation position of the inerter is controlled at both ends, with one end corresponding to the modal value ($\phi_d = 1.0$) at the mass

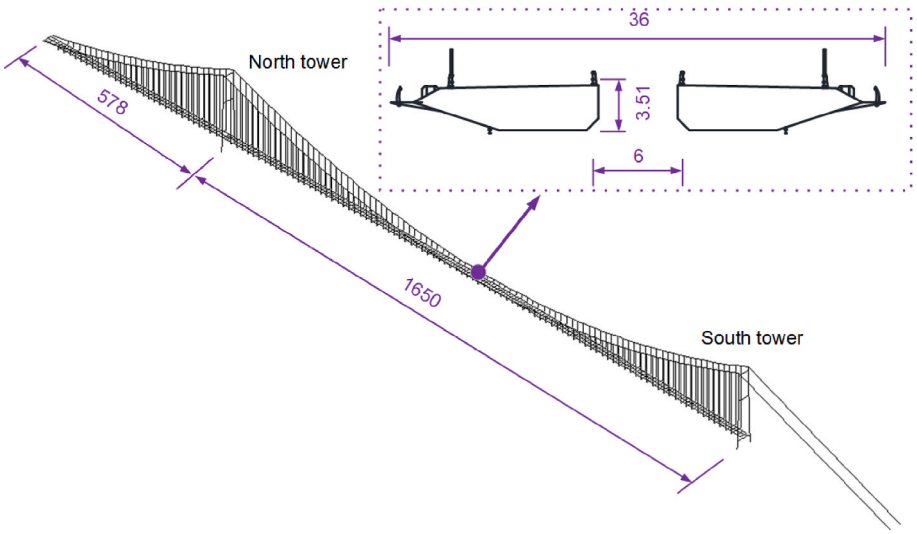


Fig. 19. Layout of Xihoumen Bridge (unit: m).

Table 1
Aerodynamic parameters affecting η_1 in VIV of long-span bridge.

Parameter	U_r	Y_1	m_r
Magnitude	[7.33 12.53] (Li et al., 2014)[6.00 11.00] (Zhang et al., 2020a) [10.26 13.84](Ge et al., 2022)	6.27 (Marra et al., 2011) 15.674 (Dai et al., 2020) [30.64 34.82] (Zhang, 2023)	1.53E3 (Zhou et al., 2017) [2.79E3 2.87E3] (Ge et al., 2022) [4.36E3 6.44E3] (Duan et al., 2022)

Table 2
Dynamic characteristics of Xihoumen Bridge.

No.	Modal frequencies (Hz)		Equivalent mass (t/m)
	Calculated values	Measured values (Li et al., 2014)	
1	0.1841	0.1831	27.29
2	0.2291	0.2295	26.75
3	0.2732	0.2734	26.64
4	0.3737	0.3625	26.49

block and the other end corresponding to the modal value (ϕ_e) at the bridge deck. The modal value ϕ_e at the bridge deck is crucial, as it indicates the spacing between the two ends of the inerter. Specifically, when ϕ_e approaches 0, it signifies a larger spacing between the two ends of the inerter, as the modal value ϕ_d at the mass block is 1, generally.

The bifurcation structure of the system with variations in S_n and ϕ_e under fixed parameters of $\eta_1 = 0.02$, $\lambda = 0.04$, $S_l = 0.01$, $\mu = 0.001$, and $\beta = 0.2$, is depicted in Fig. 17. The amplitude of LCO decreases significantly as ϕ_e decreases, especially when ϕ_e is less than 0.5, LCO disappears, as shown in Fig. 17(a) and (b). The distinction lies in the fact that as ϕ_e decreases, the phase difference increases, reaching its maximum when ϕ_e equals zero (Fig. 17(c)). The LCO amplitude is also mitigated by increasing S_n, and this effect is not influenced by changes in ϕ_e. When ϕ_e is less than 0.4, with fixed parameters of $\eta_1 = 0.02$, $S_l = 0.01$, $S_n = 10$, $\mu = 0.001$, and $\beta = 0.2$, unstable branches occur under

Table 3
Aerodynamic parameters and amplitude of bridge under VIVs.

No.	Frequency (Hz)	η_2	Y_1	ε	α_1 (y_{max}/D)
1	0.1841	17.23	77.20	1000	0.068
2	0.2291	50.48	64.74	2600	0.040
3	0.2732	79.70	57.74	3900	0.031
4	0.3737	68.11	69.29	3300	0.034

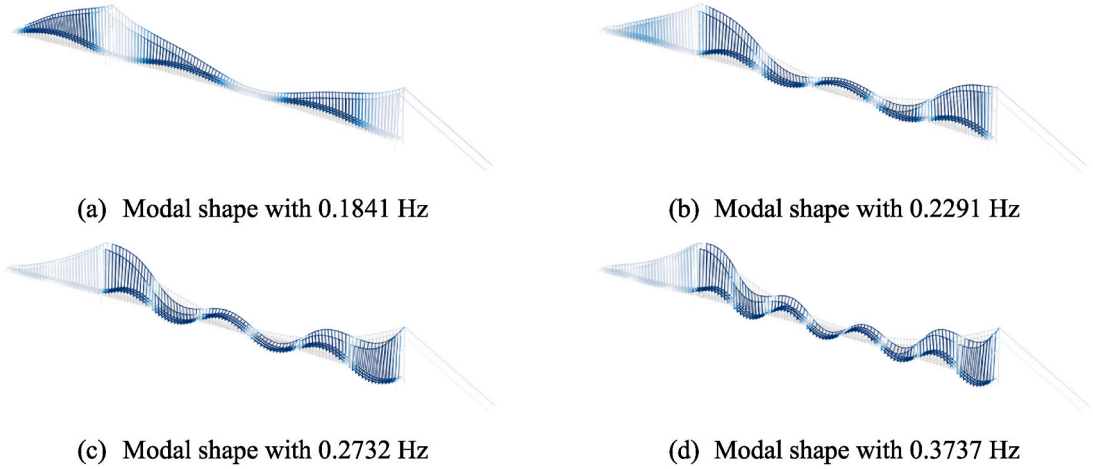


Fig. 20. Modal shapes of Xihoumen Bridge corresponding to VIV occurrence.

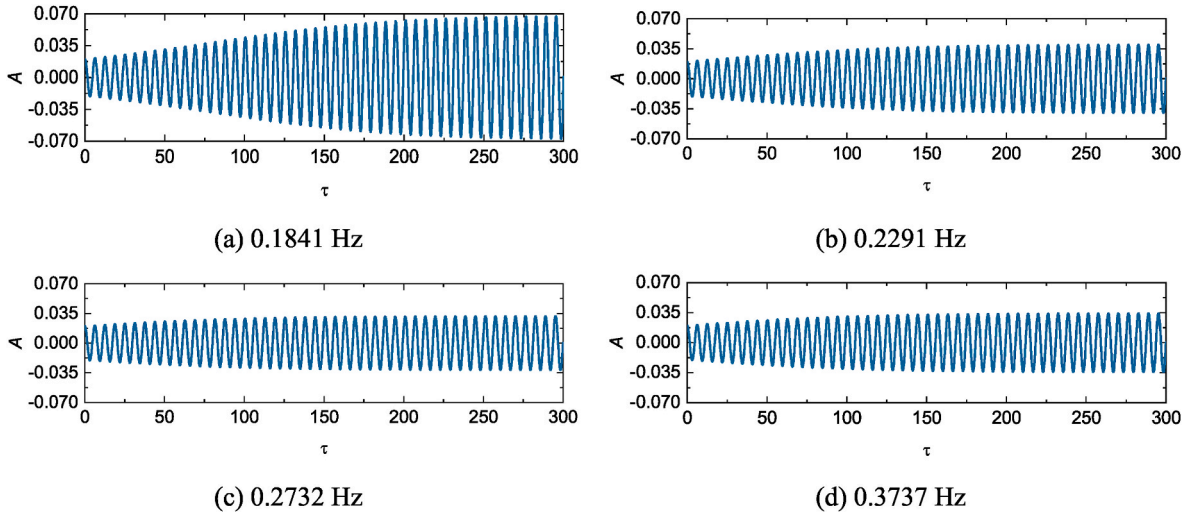


Fig. 21. VIVs of long-span bridge without control.

low mechanical damping coefficients λ ($\lambda < 0.03$) (Fig. 18(a)), resulting in LCO jump phenomena in the QZS-NESI system (Fig. 18(b)). Additionally, a small ϕ_e contributes to reducing λ required to eliminate LCO. The reduction effect on the required damping coefficient to eliminate LCO becomes insignificant once ϕ_e exceeds 0.6. The increase in the damping coefficient also results in a reduction of the phase difference, as shown in Fig. 18(c).

5. Application of QZS-NESI in VIV mitigation

5.1 Reference bridge.

The Xihoumen Bridge is a long-span suspension bridge with a main span of 1650 m and a side span of 578 m, situated in Zhoushan, Zhejiang, China. The girder of the bridge consists of twin-box girders with a width of 36 m and a height of 3.51 m. The dimensional details of the bridge are illustrated in Fig. 19. Field measurements indicate that the bridge has encountered VIVs at four modal frequencies: 0.1831 Hz, 0.2295 Hz, 0.2734 Hz, and 0.3265 Hz. The maximum VIV amplitude reached 0.2389 m at a modal frequency of 0.1831 Hz as reported by Li et al., 2014.

The ANSYS model of the bridge is constructed to reillustrate the modal dynamic characteristics associated with VIV generation. In this model, the twin-box girder and piers are simulated using BEAM4 elements, while the main cables and suspenders are simulated using LINK10 elements. The main cables are anchored at both ends and fixedly connected to the towers at the top, preventing any sliding movement. At the junction between the tower and the girder, all longitudinal constraints of the girder are released, and the lateral constraints of one end of the support are removed, while the opposite end is constrained. The dynamic characteristic of this reference bridge is obtained, and the modal characteristics associated with VIV occurrence are identified as listed in Table 2 and depicted in Fig. 20, showing good consistency with the measurement results.

5.1. VIV evolution without control

The on-site measurements show that the maximum displacements caused by the fourth-order modal VIVs are 24 cm, 14 cm, 11 cm, and 12 cm, respectively. The reduced wind speeds (U_r) corresponding to the maximum RMS amplitudes are 8.83, 10.39, 10.68, and 9.04,

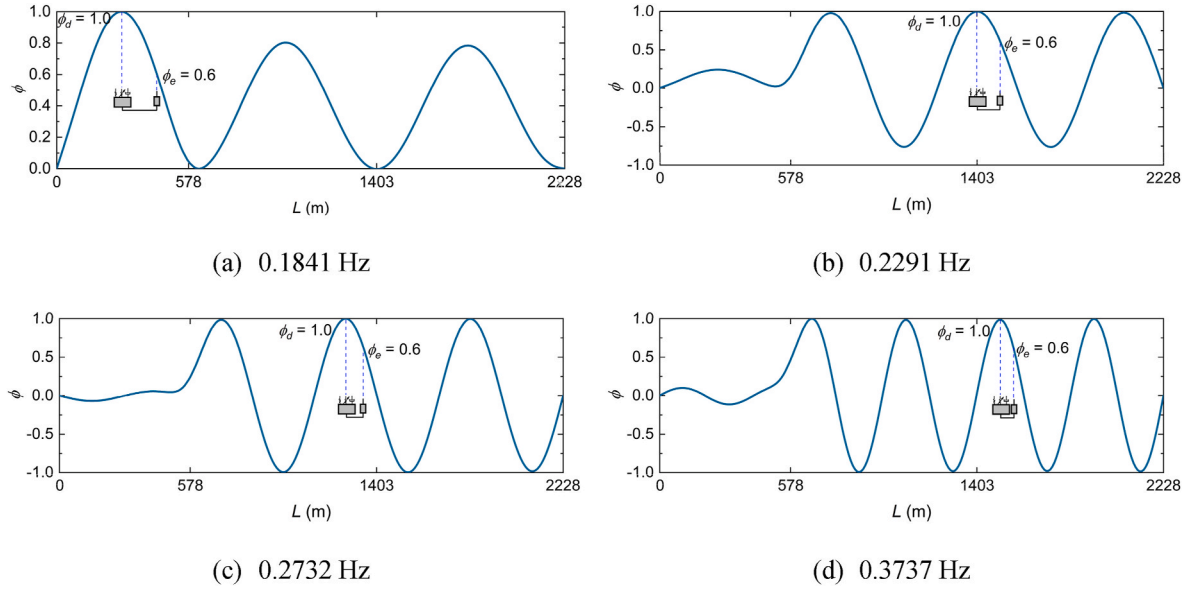


Fig. 22. Normalized mode shapes and NESI installation configuration.

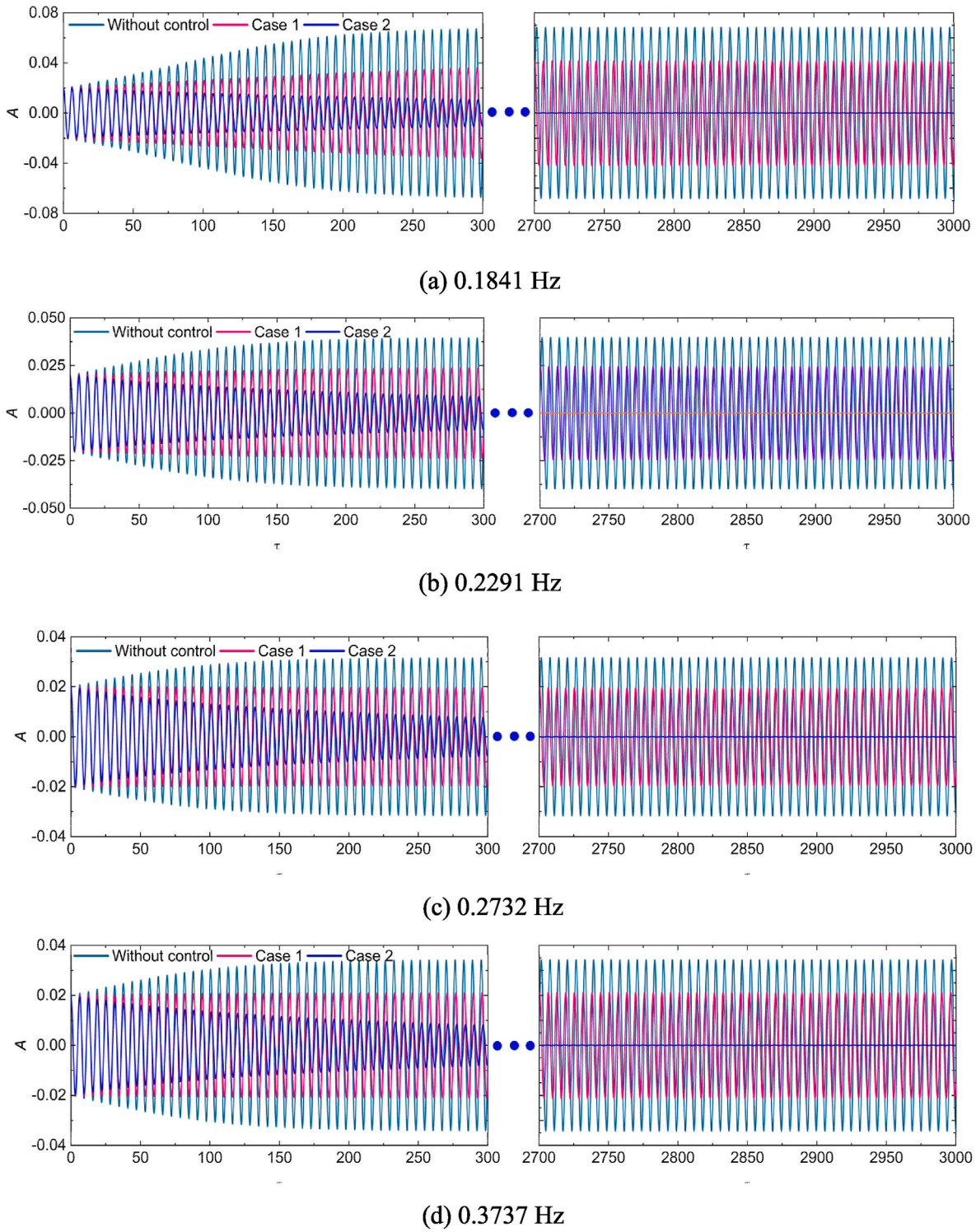


Fig. 23. VIVs of long-span bridge with QZS-NESI.

respectively. Additionally, the modal damping ratios of the fourth-order VIVs are 0.5 %, 0.51 %, 0.39 %, and 0.42 %, respectively (Li et al., 2014). Based on these parameters and the modal features of the bridge, a fixed value of $\eta_1 = 0.02$ was assumed to represent a consistent baseline vortex excitation intensity. For each mode, the nonlinear aerodynamic coefficient η_2 was determined by numerically integrating Eq. (18) under different values of η_2 , until the simulated steady-state amplitude matched the corresponding measured displacement. Subsequently, the aerodynamic coefficients Y_1 and ε were obtained by substituting the identified values of η_1 and η_2 into Eq. (20). Table 3 enumerates the dimensionless steady-state LCO amplitudes (α_1) for various modes when η_1 equals 0.02, alongside the respective VDP aerodynamic parameter η_2, Scanlan empirical coefficient Y_1, and ε. Fig. 21 depicts the VIVs corresponding to these modes, with their displacements and time scaled dimensionlessly, under initial conditions of $(A(0), \dot{A}(0)\ , V(0), \dot{V}(0)) = (0.02, 0, 0, 0)$.

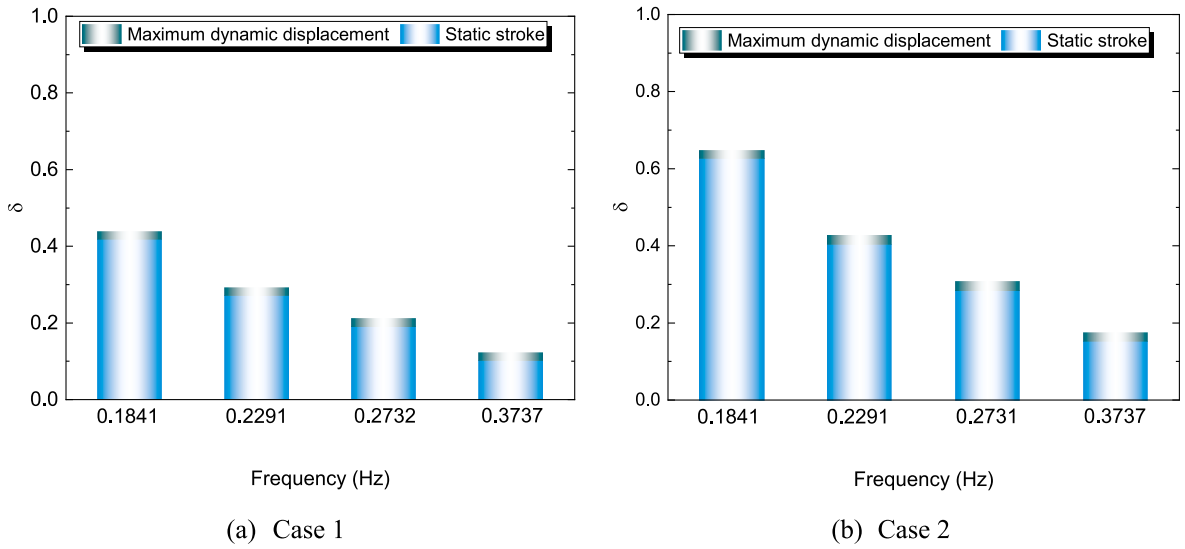


Fig. 24. Displacement of QZS-NESI under VIVs ($\alpha = 1/200$ for conservative consideration).

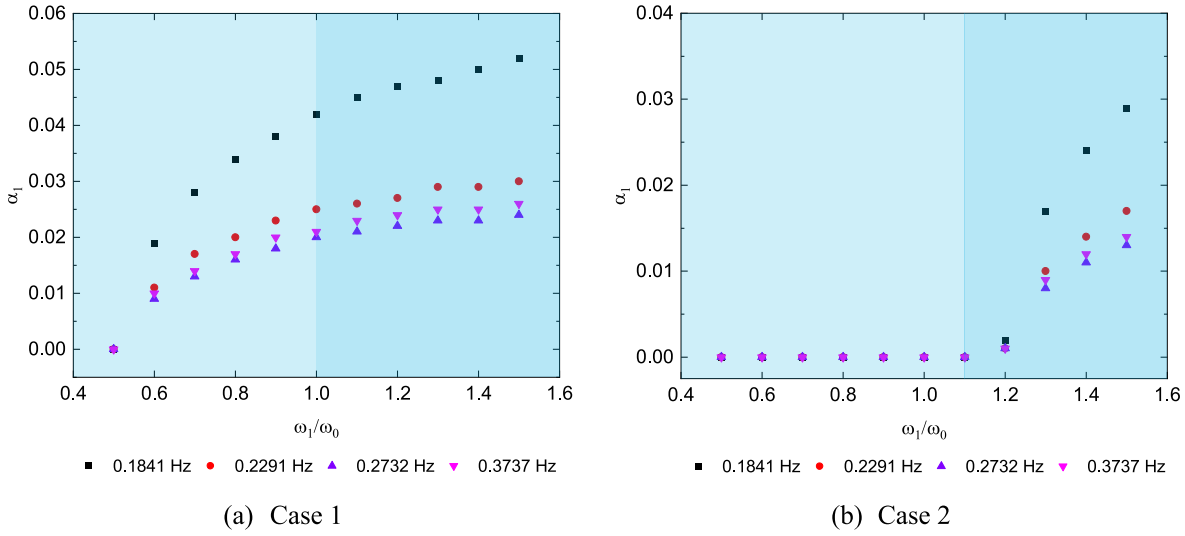


Fig. 25. VIV amplitudes under variations at each structural frequency.

5.2. VIV mitigation with QZS-NESI system

Based on the LCO bifurcation structure derived from the steady-state dynamic model, it can be demonstrated that VIV corresponding to each mode can be suppressed or even eliminated when treated independently, as illustrated in Fig. 23. To this end, two sets of parameter configurations for the QZS-NESI system are employed to achieve VIV control. The former (Case 1) is configured as $S_l = 0.01$, $S_n = 10$, $\mu = 0.001$, $\beta = 0.2$, $\lambda = 0.04$, $\phi_d = 1.0$, and $\phi_e = 0.6$, while the latter (Case 2) is configured as $S_l = 0.01$, $S_n = 10$, $\mu = 0.001$, $\beta = 0.4$, $\lambda = 0.08$, $\phi_d = 1.0$, and $\phi_e = 0.6$. For each mode, it is assumed that the NESI is installed at locations corresponding to modal values $\phi_d = 1.0$ and $\phi_e = 0.6$ with respect to that specific mode shape. This reflects a mode-specific design assumption, rather than a global placement strategy across all modes. Fig. 22 explicitly illustrates this assumed configuration. The NESI device is schematically positioned along the normalized mode shape of each VIV mode, with the mass block suspended at the peak modal point ($\phi_d = 1.0$), and the other end attached at a location where the normalized modal amplitude is 0.6. The inertance ratio and damping coefficient are doubled in Case 2 compared to Case 1, while all other parameters remain unchanged. In Case 1, VIV is suppressed (40 % reduction), while in Case 2, VIV is eliminated, which is consistent with the observations from the bifurcation results shown in Figs. 15 and 16, which demonstrates that the steady-state dynamic model derived from the CX-A method can effectively guide the VIV control of the QZS-NESI system in long-span bridges.

Fig. 24 illustrates the overall displacement of the QZS-NESI system in VIV mitigation. It can be observed that the displacement ($\delta = \Delta/D$) does not exceed the height of the girder, overcoming the challenge of large static displacements, i.e., $\Delta = g/\left(4\pi^2 f_s^2\right)$, encountered by traditional tuned mass dampers (TMDs) at low modal frequencies. For instance, at a frequency of 0.2732Hz, the static displacement of TMD is 3.39 m, representing 96.6 % of the girder height. This exceeds the allowable displacement for effective vibration mitigation. In Case 2, the displacement of the QZS-NESI system exceeds that of Case 1. This discrepancy arises from the utilization of a higher inertance ratio in Case 2 to eliminate VIV. Similar to traditional dynamic vibration absorbers, as the modal frequency decreases, the displacement of the QZS-NESI system increases. Nonetheless, it remains constrained within the height of the box girder. It is important to note that in both cases, the value of S_l was set to 0.01. Increasing S_l would lead to a further

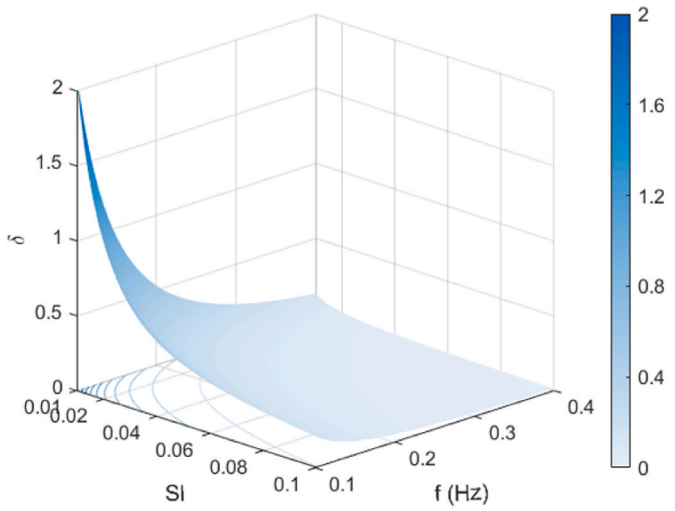


Fig. 26. Static stroke versus S_l and modal frequency ($\mu = 0.001$, $\beta = 0.4$, and $\alpha = 1/200$).

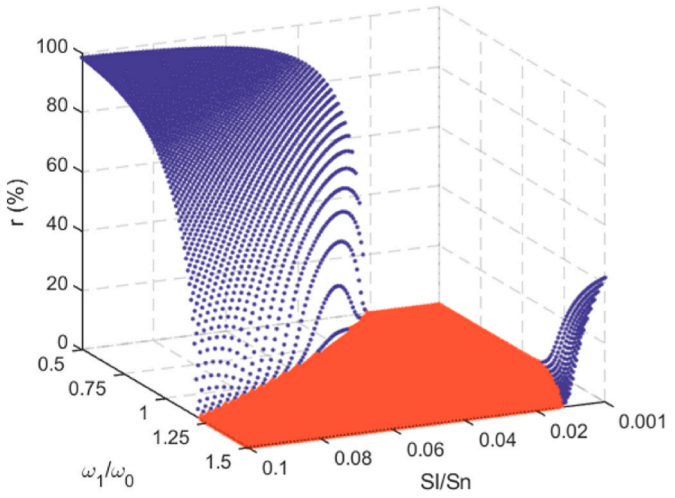


Fig. 27. VIV reduction ratio versus ω_1/ω_0 and S_l/S_n

reduction in the static displacement of the QZS-NESI system.

Fig. 25 depicts the VIV amplitude (α_1) of the primary structure under structural frequency variations of ±50 %. The QZS-NESI device demonstrates robust broadband vibration absorption characteristics, as the VIV amplitude of the primary structure can still be attenuated under frequency variations. In both cases, there exist regions of performance degradation and enhancement across various frequency ranges. In particular, the control performance of the QZS-NESI system deteriorates when the frequency ratio (ω_1/ω_0) exceeds 1, whereas it improves when the frequency ratio is below 1. This is due to the fact that, as the frequency decreases, the damping and stiffness coefficients of the QZS-NESI system increase (according to Eq. (15)), thereby leading to an enhancement in the VIV control performance. This advantageous trend can be observed in the bifurcation diagrams (Figs. 10 and 11). Conversely, the decrease in damping and stiffness coefficients caused by an increase in frequency ratio leads to a degradation in the VIV control performance of the QZS-NESI system. Nonetheless, the QZS-NESI system still manages to suppress VIV, highlighting its remarkable broad-frequency vibration absorption capability in long-span bridges.

5.3. Trade-off between static stroke and control robustness

According to Fig. 26, at the same level of VIV mitigation, the static displacement of the control system increases as the frequency decreases. To reduce static displacement in low-frequency modes, it is necessary to increase the linear stiffness coefficient S_l. However, with the nonlinear stiffness coefficient S_n remaining unchanged, this will inevitably increase the stiffness ratio (S_l/S_n), thereby weakening the nonlinear characteristics of the QZS-NESI system and subsequently diminishing its control robustness under varying frequencies. Thus, there is a trade-off between static displacement and control robustness.

Fig. 27 depicts the evolution of the VIV reduction ratio r, defined as the ratio of controlled to uncontrolled displacement, with varying frequency and stiffness ratios for the 0.1841 Hz mode under the parameters used in Case 2. The red region indicates the elimination of VIV, whereas the blue region indicates the persistence of VIV. When the stiffness ratio S_l/S_n is excessively low ($S_l/S_n < 0.01$), the VIV control performance of the QZS-NESI system degrades significantly at high frequency ratios, with a maximum reduction in the effectiveness of up to 40 %. When the frequency ratio S_l/S_n exceeds 0.02, the VIV mitigation capability of the QZS-NESI system deteriorates significantly at low stiffness ratios, with the maximum degradation reaching 100 %. The observed deterioration occurs because, when the actual frequency (ω_1) of the structure is lower than the design frequency (ω_0), the effective stiffness coefficient increases (Eq. (15)); conversely, the effective stiffness coefficient decreases. This phenomenon amplifies the decrease in linear stiffness at high stiffness ratios and the increase at low stiffness ratios. Excessively large and small linear stiffness reduces the nonlinear characteristics of the QZS-NESI system, significantly impairing its control robustness and narrowing its VIV mitigation bandwidth. According to Fig. 27, the optimal stiffness ratio range for achieving the strongest broadband VIV elimination capability is 0.01–0.02. Within this range, VIV remains fully suppressed even with a ±50 % variation in the frequency ratio.

6. Conclusions

The present study proposes a theoretical framework for the mitigation of VIVs in long-span bridges utilizing the NESI device. The steady-state dynamical models constituting this framework are derived through an approximation process of the original dynamical model, leveraging the CX-A methodology. Utilizing the steady-state dynamical models as the foundation, the bifurcation structures of the coupled system are systematically identified, taking into account variations in both external loads and design parameters. This comprehensive analysis serves as the cornerstone for the subsequent application in mitigating VIVs across multiple modes in long-span bridges, thereby providing a rigorous theoretical basis for the development of effective mitigation strategies aimed at enhancing the overall structural stability and performance. Several key conclusions that can be reached.

(1) The slow dynamics models derived from the original coupled system by the CX-A method effectively characterize the evolution of VIVs, thereby facilitating the identification of the bifurcation structures.
(2) The bifurcation structure indicates that the mechanical damping and aerodynamic damping are crucial parameters triggering unstable branches. It is recommended that the mechanical damping coefficient λ of the QZS-NESI system should not be less than 0.01 to avoid the emergence of unstable branches. There exist performance enhancement thresholds for both the mass ratio and the inertance ratio. Once these thresholds are exceeded, the improvement in the VIV mitigation capacity of the system becomes ineffective. It is advisable that the mass ratio and inertance ratio do not surpass 0.03 and 0.4, respectively.
(3) The control capability of the proposed QZS-NESI system for VIV in long-span bridge structures has been verified through the

Xihoumen Bridge. Specifically, the QZS-NESI system shows minimal static displacements when mitigating VIV at low modal frequencies, a capability that TMD cannot match. There exists a balance between static displacement and control robustness. While increased linear stiffness may lead to reduced static displacement, it substantially diminishes the nonlinear characteristics of the QZS-NESI, thereby compromising the robustness of VIV mitigation. A recommended stiffness ratio S_l/S_n of 0.01–0.02 is suggested, as it optimizes control robustness.

CRediT authorship contribution statement

Ruihong Xie: Writing – original draft, Methodology, Investigation, Data curation. **Kun Xu:** Validation, Investigation. **Peng Liu:** Investigation. **Zilong Wang:** Investigation. **Lin Zhao:** Writing – review & editing, Validation, Supervision, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A

The coefficients in slow flow dynamic models of Eqs 25 and 26are defined as

$$
\begin{aligned}
a_1 &= \frac{\frac{\beta\phi_e^2}{2} - \frac{\beta^2\phi_e^2}{2(\mu+\beta)}}{1+\beta\phi_e^2 - \frac{\beta^2\phi_e^2}{\mu+\beta}} \\
a_2 &= \frac{\eta_1}{2\left(1+\beta\phi_e^2 - \frac{\beta^2\phi_e^2}{\mu+\beta}\right)} \\
a_3 &= \frac{\eta_2}{8\left(1+\beta\phi_e^2 - \frac{\beta^2\phi_e^2}{\mu+\beta}\right)} \\
a_4 &= \frac{\left(1-\frac{\beta\phi_e}{\mu+\beta}\right)\lambda}{1+\beta\phi_e^2 - \frac{\beta^2\phi_e^2}{\mu+\beta}} \\
a_5 &= \frac{\left(1-\frac{\beta\phi_e}{\mu+\beta}\right)\frac{S_1}{2}}{1+\beta\phi_e^2 - \frac{\beta^2\phi_e^2}{\mu+\beta}} \\
a_6 &= \frac{\left(1-\frac{\beta\phi_e}{\mu+\beta}\right)\frac{3S_n}{8}}{1+\beta\phi_e^2 - \frac{\beta^2\phi_e^2}{\mu+\beta}}
\end{aligned}
\tag{A1}
$$

$$
\begin{aligned}
b_1 &= \frac{\dfrac{\mu+\beta}{2} - \dfrac{\beta^2\phi_e^2}{2\left(1+\beta\phi_e^2\right)}}{\mu+\beta-\dfrac{\beta^2\phi_e^2}{1+\beta\phi_e^2}} \\
b_2 &= \frac{\dfrac{\beta^2\phi_e^3}{2\left(1+\beta\phi_e^2\right)} - \dfrac{\beta\phi_e}{2}}{\mu+\beta-\dfrac{\beta^2\phi_e^2}{1+\beta\phi_e^2}} \\
b_3 &= \frac{\dfrac{\eta_1\beta\phi_e}{2\left(1+\beta\phi_e^2\right)}}{\mu+\beta-\dfrac{\beta^2\phi_e^2}{1+\beta\phi_e^2}} \\
b_4 &= \frac{\dfrac{\eta_2\beta\phi_e}{8\left(1+\beta\phi_e^2\right)}}{\mu+\beta-\dfrac{\beta^2\phi_e^2}{1+\beta\phi_e^2}} \\
b_5 &= \frac{\left(1-\dfrac{\beta\phi_e}{1+\beta\phi_e^2}\right)\lambda}{\mu+\beta-\dfrac{\beta^2\phi_e^2}{1+\beta\phi_e^2}} \\
b_6 &= \frac{\left(1-\dfrac{\beta\phi_e}{1+\beta\phi_e^2}\right)\dfrac{S_1}{2}}{\mu+\beta-\dfrac{\beta^2\phi_e^2}{1+\beta\phi_e^2}} \\
b_7 &= \frac{\left(1-\dfrac{\beta\phi_e}{1+\beta\phi_e^2}\right)\dfrac{3S_n}{8}}{\mu+\beta-\dfrac{\beta^2\phi_e^2}{1+\beta\phi_e^2}}
\end{aligned}
\tag{A2}
$$

Appendix B

The Jacobian matrix (J) and its elements at equilibrium points are expressed as

$$
J = \begin{bmatrix}
\dfrac{\partial f_1}{\partial \alpha_1} & \dfrac{\partial f_1}{\partial \alpha_2} & \dfrac{\partial f_1}{\partial \Phi} \\
\dfrac{\partial f_2}{\partial \alpha_1} & \dfrac{\partial f_2}{\partial \alpha_2} & \dfrac{\partial f_2}{\partial \Phi} \\
\dfrac{\partial f_3}{\partial \alpha_1} & \dfrac{\partial f_3}{\partial \alpha_2} & \dfrac{\partial f_3}{\partial \Phi}
\end{bmatrix}
\tag{B1}
$$

$$\frac{\partial f_1}{\partial \alpha_1} = a_2 - a_4 - 3a_3\alpha_1^2 - 2a_6\alpha_1\alpha_2 \sin\Phi + a_6\alpha_2^2 \sin 2\Phi$$

$$\frac{\partial f_1}{\partial \alpha_2} = a_4 \cos\Phi - \left(a_5 + a_6\alpha_1^2 + 3a_6\alpha_2^2\right)\sin\Phi + 2a_6\alpha_1\alpha_2 \sin 2\Phi$$

$$\frac{\partial f_1}{\partial \Phi} = -a_4\alpha_2 \sin\Phi - \left[a_5\alpha_2 + a_6\alpha_2\left(\alpha_1^2 + \alpha_2^2\right)\right]\cos\Phi + 2a_6\alpha_1\alpha_2^2 \cos 2\Phi$$

$$\frac{\partial f_2}{\partial \alpha_1} = \left[b_3 - 3b_4\alpha_1^2 + b_5\right]\cos\Phi + \left[b_2 + b_6 + b_7\left(3\alpha_1^2 + \alpha_2^2\right)\right]\sin\Phi - 2b_7\alpha_1\alpha_2 \sin 2\Phi$$

$$\frac{\partial f_2}{\partial \alpha_2} = -b_5 + 2b_7\alpha_1\alpha_2 \sin\Phi - b_7\alpha_1^2 \sin 2\Phi$$

$$\frac{\partial f_2}{\partial \Phi} = -\left[b_3\alpha_1 - b_4\alpha_1^3 + b_5\alpha_1\right]\sin\Phi + \left[b_2\alpha_1 + b_6\alpha_1 + b_7\alpha_1\left(\alpha_1^2 + \alpha_2^2\right)\right]\cos\Phi - 2b_7\alpha_1^2\alpha_2 \cos 2\Phi$$

$$\frac{\partial f_3}{\partial \alpha_1} = \left(a_5 - a_1 - b_6 + b_1 + 3a_6\alpha_1^2 + a_6\alpha_2^2 - 3b_7\alpha_1^2 - b_7\alpha_2^2\right)\alpha_2 + \left(2b_2\alpha_1 + 2b_6\alpha_1 - 6a_6\alpha_1\alpha_2^2 + 4b_7\alpha_1^3 + 6b_7\alpha_1\alpha_2^2\right)\cos\Phi$$
$$-\left(2b_3\alpha_1 - 4b_4\alpha_1^3 + 2b_5\alpha_1\right)\sin\Phi + \left(2a_6\alpha_2^3 - 6b_7\alpha_1^2\alpha_2\right)\cos^2\Phi$$

$$\frac{\partial f_3}{\partial \alpha_2} = \left(a_5 - a_1 - b_6 + b_1 + a_6\alpha_1^2 + 3a_6\alpha_2^2 - b_7\alpha_1^2 - 3b_7\alpha_2^2\right)\alpha_1 - \left(2a_5\alpha_2 + 6a_6\alpha_1^2\alpha_2 + 4a_6\alpha_2^3 - 6b_7\alpha_1^2\alpha_2\right)\cos\Phi - 2a_4\alpha_2 \sin\Phi + \left(6a_6\alpha_1\alpha_2^2 - 2b_7\alpha_1^3\right)\cos^2\Phi$$

$$\frac{\partial f_3}{\partial \Phi} = -\left(b_2\alpha_1^2 + b_6\alpha_1^2 - a_5\alpha_2^2 - 3a_6\alpha_1^2\alpha_2^2 - a_6\alpha_2^4 + b_7\alpha_1^4 + 3b_7\alpha_1^2\alpha_2^2\right)\sin\Phi - \left(a_4\alpha_2^2 + b_3\alpha_1^2 - b_4\alpha_1^4 + b_5\alpha_1^2\right)\cos\Phi - \left(2a_6\alpha_1\alpha_2^3 - 2b_7\alpha_1^3\alpha_2\right)\sin 2\Phi \tag{B2}$$

Data availability

Data will be made available on request.

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